

Master Thesis

Syntax versus Semantics:
Adapting the Microportraits Pipeline with Semantic Role Labeling

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Abstract

The subjectivity of constituting what a stereotype is, makes stereotype detection a difficult task within Natural Language Processing (NLP). Designing annotation tasks to create datasets for training machine learning models becomes challenging to avoid subjective annotations. The Microportrait Pipeline avoids this subjectivity by extracting summaries about target entities directly from natural language itself (Fokkens et al., 2018). The use case for developing the pipeline is to compare how Muslims are represented in Dutch news articles compared to Dutch people. Each portrait of an entity contains the labels that are used to refer to a social category, the properties that describe an entity and their roles and behaviors. Biases can occur when speakers choose such labels, properties and roles to describe a certain entity or social group (Beukeboom and Burgers, 2019), which can come to the surface through the patterns within the extracted microportraits. However, the roles and behaviors as they are extracted in the original design of the Microportrait Pipeline are based on syntactic dependencies, which we substitute in this study with a semantic role labeling (SRL) system for Dutch.

To do this, we finetuned an XLM-RoBERTa model with Dutch and English data from the Universal Propositions bank to supplement the pipeline with SRL. The model has been evaluated using standard metrics, and yielded F1 scores for the agent and patient roles above 0.80. The SRL information is included in the Microportrait Pipeline to be able to account for the semantic roles predicted by the SRL model. This information should provide more accurate semantic representations of the roles and behaviors assigned to specific entities, as well as create microportraits that contain more semantic associations rather than syntactic associations. The output of the SRL-based pipeline and the syntax-based pipeline are compared on a quantitative and qualitative level. The quantitative level explores PMI scores with a minimum frequency threshold to investigate how the associations with target words changes between the syntax- and semantics-based portraits. The qualitative level evaluates how the quality of the extracted portraits changes in terms of how informative the extracted portraits are, as well as how the assigned roles and behaviors change. This is done by comparing both outputs to a manually annotated gold subset.

The results show how including SRL changes the microportraits it outputs. Including SRL in the Microportrait Pipeline overrides quite some heuristics in the original pipeline, which leads to a more accurate extraction of roles and behaviors in the SRL portraits. The pipeline based on SRL is better able to generalize in making its portraits, which leads to less noise in the eventual microportraits, yet comes with the trade-off that it currently still omits some information. This study shows a promising improvement of the Microportrait Pipeline to be able to reason about stereotypical views as they occur in natural language, as well as propose further suggestions to improve the SRL-version of the pipeline.

Declaration of Authorship

I, Sanne van der Wal, declare that this thesis, titled *Syntax versus Semantics: Adapting the Microportraits Pipeline with Semantic Role Labeling* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: 24/06/2026

Signed: 

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Chapter 1

Introduction

Stereotypes are a fact of life: different cultural (sub)groups can have stereotypical beliefs to make sense of their surroundings and other groups, politicians can use stereotypes to create scapegoats, feed into the polarizing idea of *us versus them*, or to place their political opponents into these stereotypic boxes (e.g., the leftist elite). Yet, not only politics or media are riddled with stereotypes, in advertising they know how to use stereotypes to cater towards men by focusing on grilling essentials, or sell new beauty products to women. This man-woman gender division is fed with stereotypical beliefs. Not only does it transcend cultures and goals, stereotypes exist within different topics as well. For example, climate activists are portrayed as hippies, immigrants are associated with luck-seekers and asylum seekers are put in the box of terrorists. We are constantly being exposed to stereotypic views, whether we notice them or not, whether we agree with them or not, they are a mechanism to help make sense of the world. However, stereotypes are not just harmless ways of representing certain groups of people, they can influence thoughts and behaviors and instigate discrimination and violence. Knowing what stereotypes are, how they reside within our language and what effects they have is then not trivial information, but vital knowledge.

This thesis aims to further develop a computational linguistic approach that aims to retrieve information from language that makes it possible to reason about stereotypical beliefs. This approach is called the Microportrait Pipeline, which extracts summaries about entities that are mentioned in a text. For example, an article about immigration can contain the following sentence, which is made up for illustration purposes:

> *Violent immigrants infiltrate our society.*

This sentence contains two entities: *immigrants* and *society*. The portrait of *immigrants* would contain their property of being violent, as well as their position in the sentence, namely that they are the `agent` for the verb *infiltrate*. The portrait for *society* would contain the property that it is *our* society, as well as its role of `undergoer`, since it undergoes the event of infiltrating. However, the creation of these portraits, especially when assigning the roles an entity plays for a verb, is now based solely on syntactic information. The assigned roles then answer the question: *which syntactic relation does the entity have to the verb?* The goal is that the system that assigns these roles does this by answering the question: *who did what to whom?* As such, the information that it would provide is much more focused on the meaning of a verb and the participants that it requires, rather than on the syntactic relations that connect a word to a verb.

This thesis aims to create a system that is able to provide an answer to the question *who did what to whom?* As such, the portraits that are extracted by the Microportrait Pipeline are much more rooted in semantics, rather than only being based on syntactic structures. This should result in an updated Microportrait Pipeline that extracts more accurate summaries about entities within a text, which would result in more accurate reasoning about any patterns that show up across the extracted microportraits.

1.1 Research Aims

As such, the goal of this thesis is to evaluate how the extraction of microportraits improve when semantic information is added to the currently syntax-based pipeline. To guide the research design of this thesis, the following research question has been posed:

> *To what extent can the inclusion of semantic role labeling in the microportrait extraction pipeline increase the granularity and improve the accuracy of microportrait extraction in Dutch News?*

This question is investigated by creating a Semantic Role Labeling system for Dutch and implementing this in the Microportrait Pipeline. The evaluation happens on both a quantitative and qualitative level. This shows that by basing the information on semantics, the pipeline creates less noise within its portraits, and overall shows more accuracy in the extracted roles. These are promising results for the improvement of the Microportrait Pipeline.

1.2 Outline

The rest of this thesis is structured to take you along the road of improving the Microportrait Pipeline. First, Chapter 2 will introduce the social psychological grounding of stereotypes, how stereotypes function within language, what NLP research has already been done to detect stereotypes, as well as introduce microportraits and the theoretical basis in linguistics for improving the pipeline with semantic role labeling. Chapter 3 explicates the steps that have been undertaken to realize the implementation of semantic information into the pipeline. The results of the evaluation can be found in Chapter 4. This chapter shows both the results of the quantitative and qualitative evaluation, which are then further interpreted for its implications in the discussion of Chapter 5. Finally, concluding remarks are made in Chapter 6.

Chapter 2

Theoretical Background

2.1 On Stereotypes

This section explores what stereotypes are, especially as set out in the social psychological literature. When researching stereotypes, especially when wanting to create a sound framework to extract them from natural language, it is vital to know what they are, how they function and how language contributes to the existence of stereotypes. First, the notion of social categories is discussed, followed by a section how language functions as a powerful tool to construct, maintain and reflect certain beliefs.

2.1.1 Social Categories

Stereotypes are a product of a process known as categorization, which is a process that helps in understanding a thing by being able to relate it to another thing, or to be able to say what it differs from (Oakes, 2003). As such, it is with the help of social categorization that we are able to identify things, or entities, and be able to situate them within our understanding of a social context (Oakes, 2003). However, not every category is attributed negative characteristics in every social context, yet can simply function within the human cognition as a neutral way to make sense of the world. The process of categorization is a part of human and social life, which functions to align the individual with the current social context (Oakes, 2003). Within social psychology, social categorization has been argued to be a process rooted in intergroup bias, where an individual can be reduced to a certain category (Beukeboom and Burgers, 2019). It follows that the perception of an entity or situation can then differ from reality, and create efficient, yet inaccurate interpretations of a social situation (Oakes, 2003). Once a category of a social group is learned, it is then possible to attribute characteristics to that social group, which is also known as forming a stereotype (Bar-Tal, 1996). Where the features of a social category are represented by stereotypes, stereotypes give rise to prejudice, which in turn gives rise to discrimination (Bodenhausen and Richeson, 2010; Bodenhausen and Macrae, 2013). When antipathy towards a social group is developed based on a categories' features (prejudice), it can lead to people treating the members of that social group disadvantageously (discrimination) (Bodenhausen and Richeson, 2010). As such, stereotypes pave the way for prejudice and discrimination, which are the next steps in the so-called pyramid of hate (Cignarella et al., 2026). Stereotypes are acts of bias, which form the basis for acts of prejudice, acts of discrimination, bias-motivated violence and, ultimately, genocide. The representation of an entity or social

situation in the mind of a perceiver is one thing, yet how it then works to influence the perceiver's overt behavior is another (Bodenhausen and Macrae, 2013). Mental representations can affect overt behavior, and understanding how these representations are constructed and maintained is vital to control them. That is also why even when negative characteristics are attributed to a category, Operario and Fiske (2003) argue that despite stereotypes being inherent to human nature, they can be controlled with personal motivation and effort.

Given this ambivalence of stereotypes, the effects of the positive or negative aspects of a stereotype are largely determined by the social context in which they arrive (Operario and Fiske, 2003). Operario and Fiske (2003) states that stereotypes can be both (a) basic human tendencies that are inherent to human cognition and (b) potentially damaging and harmful belief systems, depending on the social context and the power structures at play in a certain situation. As such, as the media often reports on entities and different social groups, it should be acknowledged that in largely multi-cultural societies where members of different groups may have little interaction with each other, it is via the media that may be the only way they learn about each other (Kidd, 2016). This leaves the media with a powerful position, one that can make or break the chain of stereotype, prejudice and discrimination.

2.1.2 Power of Language

For a stereotype to be communicated among people, language can be seen as the default form to share such beliefs. Lippmann, who introduced the term stereotypes in 1922, refers to them as"pictures in our head" (Lippmann, 1922). The idea that stereotypes exist within human cognition as mental representations also lies at the base of a theory of meaning. This can be further illustrated by the linguistic theory of Ogden and Richards (Ogden and Richards, 1930), which states that the meaning of a word should be seen as tripartite. This triangle of meaning, or semantic triangle, puts words, thoughts and things at each corner, of which a simplified representation can be found in figure 2.1.

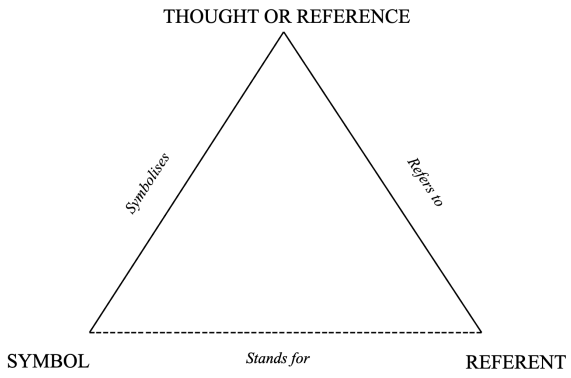


Figure 2.1: Semantic triangle

The most relevant connection in the triangle is that between the symbol, words and utterances, and thought, the mental images in our heads (often referred to as

reference). It is argued that the relation between the symbol and the referent, the thing it refers to in the real world or referential meaning, does not exist and is merely an arbitrary relation. The relation between the symbol and the reference, on the other hand, is causal. This means that the chosen symbolism during a speech situation is caused partly by the reference that is being made and partly by social and psychological factors (Ogden and Richards, 1930). That is, the symbol or word that has been chosen, is chosen with the purpose of having an effect on other people, as well as the speaker trying to convey their own attitudes (Ogden and Richards, 1930). As a result, the chosen symbols result in an act of reference on the hearer's part (Ogden and Richards, 1930). This act of reference where the symbol is being related to a thought is then followed by assuming an attitude (Ogden and Richards, 1930), which can be influenced by the social or cultural (sub)group that the hearer is a part of. The words that are chosen by a speaker to communicate about an event or entity then cause how a hearer processes what is being communicated. The form then influences the process on how a thought about the event or entity is formed, based on the mental representations that are evoked by it. This shows the link between the mental image and the form of language, or how language in its arbitrary form functions to share beliefs and attitudes - including stereotypes.

Words, and all other aspects of language, are ubiquitous. Everywhere around us we can find a form of language that attempts to communicate something and which triggers the receiver to perform these acts of reference to arrive at a mental representation or thought. The words that are chosen, and how they are interpreted is then rooted in these social and psychological factors, which is what makes language a powerful vessel. How we choose to use language is capable of shaping society, yet those choices themselves are also shaped by society. It is a tool that helps us reflect, construct and maintain certain beliefs (Beukeboom and Burgers, 2019). These beliefs do not exist on their own or within one individual, members of a cultural (sub)group share these mental representations, organized into social attitudes (Van Dijk, 1995). Through sharing these representations, groups are able to establish an identity, norms and values among their members (Van Dijk, 1995). Attitudes, beliefs and as such stereotypes then become a product of collections of individuals, as these views are consensually shared among them (Beukeboom and Burgers, 2019). Being part of a certain culture means that members have internalized shared stereotypes and consequently create a shared social reality (Beukeboom and Burgers, 2019).

Now that we know how thought both influences language choices and affect how we interpret language choices, social and cultural situatedness impacts how we structure our thoughts. Language and communication are at the center of these processes, since it is via interpersonal communication, and mass media that certain representations are shared and maintained (Beukeboom and Burgers, 2019). Nonetheless, language does not only exist to maintain and normalize certain beliefs, it is also a tool that can resist the status quo Van Dijk (2015). As such, the power of language entails two sides of the same coin, and knowing how they occur, are created, maintained and communicated can help to change stereotypic representations.

2.2 The Linguistics of Stereotypes

The section above explicated how social categories are theorized within social psychology, as well as the theoretical background on how language functions to birth and keep

stereotypes alive. Advancing from this abstract topic, this section will discuss literature that has researched how stereotypical beliefs exist in language, and in what ways (computational) linguistic methods help to advance this research topic.

Finally, this section will introduce the Microportrait Pipeline and motivate how this thesis aims to further develop its architecture.

2.2.1 Stereotypes in NLP

As discussed above, language and stereotypes are inherently connected. Language is the vessel that communicates stereotypic representations that exist within human cognition, and the computational linguistics provides the tools to extract those representations from natural language. Within the field of Natural Language Processing (NLP), much research has been done on extracting harmful language from text that feeds into stereotypical beliefs, yet the field of stereotype detection has only been in fruition since 2019 (Cignarella et al., 2026). Nonetheless, NLP does have a rich history of extracting explicit harmful language from natural language data, such as hate speech detection which already required surveys in 2017 (Schmidt and Wiegand, 2017). However, stereotype detection focuses more on the implicit layers of language, since stereotypes are not always overt, offensive utterances, but become normalized and accepted within certain subcultures. This section will further discuss some methods that the NLP research community has already put forward to detect stereotypes.

For example, Mendelsohn et al. (2020) propose a framework for the computational linguistic analysis of dehumanization. By drawing upon social psychology research, dehumanizing language is identified through linguistic aspects of salient components of dehumanization (Mendelsohn et al., 2020). One salient component of dehumanization is how stereotypes function within language and, as it happens, research has shown that word embeddings capture stereotypical views that exist in the real world (Bolukbasi et al., 2016; Manzini et al., 2019). Based on this knowledge, Mendelsohn et al. (2020) exploit this inherent bias of word embeddings by quantifying stereotypes about LGBTQ+ people in news data. They create vector representations for keywords of interest, such as *gay*, which makes it possible to identify its nearest neighbors via cosine similarity. Since word embeddings are vectors in space, cosine similarity makes it possible to calculate the angle between those vectors in order to establish their similarity. Its most similar words, or nearest neighbours, create a picture that shows which words are most closely related to a specific keyword embedding (Mendelsohn et al., 2020). The results indicate how language surrounding gay people becomes less dehumanizing throughout time, by showing how its closest associated words change in connotation throughout time (Mendelsohn et al., 2020). Furthermore, quantifying stereotypes through word embeddings throughout a set period of time can bring historical trends and social change to light (Garg et al., 2018). Word embeddings provide a powerful tool that can help in the development of metrics to characterize stereotypes and attitudes, especially by tracking the evolution over time (Garg et al., 2018).

Furthermore, efforts have been made to quantify stereotypes not through embeddings, but by creating a numerical annotation task (Liu, 2024). Liu (2024) decided to develop a continuous variable scale to identify stereotypes, rather than a binary stereotype versus. non-stereotype task. The annotation task of stereotypes is highly subjective, which makes it difficult to realize. Liu (2024) overcame this by using a Best-Worst-Scaling (BWS) scheme to account for this complexity of the task. That means that annotators should identify the most and least stereotypical sentences in

a quaternion - a tuple of four sentences (Liu, 2024). The scores for the highest and lowest stereotypes within a tuple are then converted to a value between -1 and 1. This annotation task can then show which of the bias types has the highest scores for stereotypical language. The bias types refer to stereotypes for professions, race, gender, and religion. This annotated dataset is then used to train LMs such as BERT, DistilBERT and RoBERTa to predict stereotypes scores for these bias types in hate speech (Liu, 2024). The reason for making this a manual task is to be able to evaluate bias that exists within the embeddings of masked language models (MLMs), and subsequently debias them (Liu, 2024).

Additionally, Schmeisser-Nieto et al. (2022) have developed criteria for the annotation of implicit stereotypes. This task annotated sentences with a binary value whether it held a stereotype or not, as well as whether the stereotype is then communicated implicitly or explicitly. The focus of the annotation task is to decide whether the stereotype is communicated implicitly, which results in an annotation scheme that can help in this decision. The scheme exists of 13 non-mutually exclusive binary categories that can be identified within a sentence. For example, the existence of anaphora, rhetorical questions, the need for world knowledge to infer the stereotype, irony, metaphors and jokes all contribute to a stereotype being communicated implicitly (Schmeisser-Nieto et al., 2022). This is mostly based on the steps of inference an interpreter has to undertake in order to arrive at a stereotype, which makes it both a highly difficult annotation task as well as an important one. The inter-annotator agreement showed high variation among the 13 criteria, which underlines both the difficulty of establishing such implicit criteria, as well as the subjectivity that comes with such a task. The authors propose different ways of improving the annotation scheme and acknowledge the challenges that arise with such a task (Schmeisser-Nieto et al., 2022). Nonetheless, eventually such schemes would be applied on machine learning models to improve the automatic identification of stereotypes as a classification task, and the errors of machine learning models can also provide information on which types of implicit stereotypes remain difficult to categorize (Schmeisser-Nieto et al., 2022).

Quantifying stereotypes through embeddings or annotation schemes do come with their limitations. First of all, embeddings are robust representations that offer a new way of conducting applied cultural and social scientific research related to meaning (Arseniev-Koehler, 2024; Rodriguez and Spirling, 2022). However, limitations arise when using static embeddings that do not accurately capture polysemy and struggle with out-of-vocabulary (OOV) words. This means it would make more sense to use contextual embeddings, yet this comes with the black-box issue of interpretability as well as computational complexity (Alkaabi et al., 2025). Moreover, when calculating the similarity between embeddings, cosine similarity may not always be the most promising metric to use (Steck et al., 2024). Lastly, embeddings focus on the distributional hypothesis of meaning, and as such does include semantics in its study of stereotype, yet does not include any syntactic properties of language. As for annotation schemes, this of course comes with difficulty of having annotators with their own subjectivity and creating an annotation task that is as neutral as possible. In addition to this challenge, manual annotations are highly time-consuming when creating vast amounts of datasets for training machine learning models.

In conclusion, the current NLP approaches for stereotype detection include embeddings, annotation schemes and classification tasks. They each have their merits for conducting stereotype detection, yet also come with a lack of interpretability for

embeddings, and the difficulty of dealing with subjectivity for annotation tasks. The following section will introduce the Microportrait Pipeline that aims to mitigate both issues.

2.2.2 The SCSC framework and Microportraits

The Microportrait extraction pipeline aims to extract an impression of a target group or individual from a single text (Fokkens et al., 2018). These impressions are taken directly from the language itself, extracting nouns, adjectives and semantic roles. As such, it aims to provide a means to reason about stereotypical beliefs within language by extracting the forms and constructions directly from language itself. How language itself functions to communicate stereotypes is captured in the Social Categories and Stereotypes Communication (SCSC) framework (Beukeboom and Burgers, 2019). Beukeboom and Burgers (2019) base their work on the different forms of linguistic bias they established in an earlier paper (Beukeboom and Burgers, 2017), as well as on social psychology by establishing three variables that are formed and maintained through specific aspects of language. Namely that of 1) perceived category entitativity, 2) cognitive stereotype content and 3) perceived category essentialism (Beukeboom and Burgers, 2019). Perceived category entitativity refers to what extent a social group is considered to be a coherent, unified and meaningful entity as opposed to a loose set of individuals (Beukeboom and Burgers, 2019). This relates to the perceptual features that can tie a group together, as well as invisible commonalities, such as ancestry or common origin. One example of perceived category entitativity within language can be by referring to a whole group of people by the fact that they wear hijabs, or using a derogatory term for it such as the *kopvoddentaks* (head rag tax) proposed by the Dutch politician Geert Wilders. The second variable of cognitive stereotype content refers to the cognitive representation people can hold about a social group, which consists of beliefs, expectancies about behaviors, features, and traits (Beukeboom and Burgers, 2019). This can be recognized in language by assigning a certain group a constant role within in events. If we were to ask the question *who did what to whom* about a certain situation, one social category would always come out as *they did that to them/us*. Lastly, perceived category essentialism refers to a set of characteristics that is perceived to be immutable to its members (Beukeboom and Burgers, 2019). This essentialism is reflected by the ability to make rich inferences about the members of a social group, every member of a group in whatever situation has those characteristics. To recognize such essentialism in language, it is worthwile to extract patterns about how category entitativity and stereotype content are structurally reflected within language.

The SCSC framework is what inspired the Microportrait extraction pipeline (Fokkens et al., 2018). By creating these impressions about individuals or target groups from a single text can then be combined on a larger scale that can reflect how they are perhaps systematically portrayed within the media - or any other type of platform. The pipeline extracts three different types of information:

Labels Biases exist in the label content that is chosen to refer to a category of people in the real world. This label communicates which social category is being referred to, and simultaneously conveys the position of the category in a hierarchical social-category taxonomy (Beukeboom and Burgers, 2019). This category is concerned with the nouns that are chosen to convey meaning about a category and its members.

Properties A different linguistic form of label can be found in adjectives, or words that describe a given entity. Adjectives refer to properties of objects and to subordinate level distinctions (Beukeboom and Burgers, 2019). As a result, entities can often possess many qualities as described by adjectives, yet only be positioned in one category as described by a noun (Beukeboom and Burgers, 2019).

Roles Lastly, biases also exist in describing behaviors of categorized individuals. A set of stereotypic characteristics can be assigned to a category by *how* information is formulated. As such, formulations refer to what is expected for the categorized individual or applied category as a whole (Beukeboom and Burgers, 2019). The Microportrait Pipeline investigates this by extracting which roles categorized individuals play in events, where an event constitutes a predicate.

Labels, properties and roles fall under the term descriptions; the basic unit of a microportrait. To illustrate how the pipeline works to extract descriptions form text, Fokkens et al. (2018) provide an example sentence of *the pious Muslim smiled*. This sentence contains the label *Muslim*, the property *pious* and the agent or ARG0 role for *Muslim* in the event of *smiling*. The microportrait of a social category or entity is the collection of all these descriptions within a certain text, providing so to say a 'summary' about how they are represented. Microportraits on their own cannot provide information about the stereotypical characteristics assigned to a social group or entity. It is only by investigating patterns in the extracted microportraits that biases can be identified in which information is communicated about a social group, how it is communicated and to what extent it differs from other groups (Fokkens et al., 2018). This comparison can be made by calculating the Pointwise-Mutual Information score of descriptions that occur in the same microportrait to see which descriptions are typically used for one group compared to the other (Fokkens et al., 2018).

Different from predefined annotation tasks, microportraits are *stereotype agnostic*. Relying on manually constructed annotation schemes that are tailored to a specific target entity, such as gender stereotypes, immigrant stereotypes or for stereotypes about profession, requires researchers to define how stereotypical beliefs should be categorized beforehand, as well as decide which are relevant for the task at hand. As a result, this limits the research by the beliefs encoded in the annotation scheme and may overlook any stereotypical associations that fall outside the predefined annotation task.

Microportraits then take a fundamentally different approach. Rather than starting from preconceived notions about which stereotypes exist, they extract descriptions directly from language in the form of labels, properties and roles. Because the extraction process is independent of any particular target group or stereotype category, the same methodology can be applied to any social group, individual or entity. That is, microportraits are target-agnostic and allow stereotypical associations to emerge from the data itself instead of being defined beforehand. By comparing the descriptions that systematically co-occur with different social groups, it is possible to identify recurring patterns about a group without requiring prior knowledge of the stereotypes that may be present. Similar to embedding-based approaches, microportraits do not depend on manually specified stereotype categories. However, unlike embeddings, microportraits are transparent about the linguistic structures through which stereotypes are communicated. They provide interpretable descriptions in terms of labels, properties and semantic roles. As a result, microportraits not only provide a method for the detection of stereotypical associations, but also provide insight into how these associations are

communicated through language.

2.2.3 Roles and Behaviours

Following the SCSC framework, the Microportrait Pipeline follows the organization of labels, properties, and roles to create stereotype-like summaries. Labels and properties are extracted by identifying nouns and its modifiers through dependency relations, respectively. The current extraction of the behaviors is also extracted using dependency relations, where the subject of an active sentence is an `agent` and of a passive sentence the `undergoer`, the object of an active sentence is an `undergoer` and an indirect object depending on the verb a `recipient`. Despite these rules capturing many of the semantic structures that can be derived from its syntax through a few simple rules, it does choose some shortcuts that do not capture more complicated semantic structures. The current approach then comes with two issues: 1) the choice of semantic roles and 2) the rule-based system that assigns these roles.

The first issue is the use of the three semantic roles `agent`, `undergoer` and `recipient`. These three roles often play the biggest roles in sentence structures, yet they do not capture the wide variety of the different participants an event can have. The most standard set of thematic roles can be found in table 2.1.

Role	**Description**
AGENT	Initiator of some action, capable of acting with volition.
PATIENT	entity undergoing the effect of some action, undergoing some change in state.
THEME	the entity which is moved by an action, or whose location is described.
EXPERIENCER	the entity which is aware of the action or state described by the predicate, but which is not in control of the action or state.
BENEFICIARY	the entity for whose benefit the action was performed.
INSTRUMENT	the means by which an action is performed or something comes about.
LOCATION	the place in which something is situated or takes place.
GOAL	the entity towards something moves, either literally or metaphorically.
SOURCE	the entity from which something moves, either literally or metaphorically.

Table 2.1: Thematic roles and their descriptions (Saeed, 2015)

Table 2.1 shows the different thematic (or semantic, used interchangeably in this thesis) roles that can be identified for an event. However, defining these semantic roles in general remains a challenge, namely what semantic basis exists for characterizing these roles (Saeed, 2015)? To avoid the blurred lines between some semantic roles, Dowty proposes a solution where semantic roles are defined in terms of entailments of the predicate (Dowty, 1991). By viewing roles as a cluster of entailments about an argument position which are shared by some verbs, Dowty shows how to move away from bounded categories and see thematic roles as prototypes (Dowty, 1991), with varying degrees of membership. He proposes a Proto-Agent role and a Proto-Patient role, of which each has the following entailments as defined by Dowty (1991):

2.1 Properties of the Proto-Agent role:

a. volitional involvement in the event or state

b. sentience (and/or perception)

c. causing an event or change of state in another participant

d. movement (relative to the position of another participant)

2.2 Properties of the Proto-Patient role:

a. undergoes a change of state

b. incremental theme

c. causally affected by another participant

d. stationary relative to movement of another participant

As such, these prototypical roles provide some more flexibility in assigning semantic roles to sentence participants, where those participants that appear in one argument position can all meet some entailments that are prescribed by the verb. This notion is further discussed later on.

The second issue with the current pipeline is the rule-based system that assigns the semantic roles. That is, the assumption that a grammatical relation defines the semantic role. However, the subject position in an active sentence is not always an agentive participant. An example taken from Saeed (2015) of how the subject position in almost exactly the same sentence takes on two different semantic roles can be found in sentence 2.3 and 2.4 below:

2.3 John took the book from Bill in order to read it.

2.4 John received the book from Bill in order to read it.

This shows how *John* in 2.3 assumes an agent position, whereas *John* in 2.4 is a recipient. Both sentences remain active in voice, yet a change of the semantics of the verb dictates which roles can be token on by whom in the sentence.

Additionally, a change in syntax allows a speaker to change the semantics of the sentence, by omitting participants through assigning other participants a different position. Another example is shown in sentences 2.5 -2.7 (Saeed, 2015):

2.5 Ursula broke the ice with a pickaxe.

2.6 The pickaxe broke the ice.

2.7 The ice broke.

These examples show how *Ursula* takes on both the subject and agent position in 2.5, *the ice* is the direct object and patient, and *the pickaxe* is in a prepositional phrase that takes on the instrument role. Sentence 2.6 omits the agent, and fronts the instrument into the subject position. Sentence 2.7 shows how both the instrument and agent are removed from the sentence, and that the verb *break* then allows the *the ice* as patient to take on the subject position. Nonetheless, this example shows how the subject position in an active sentence is not always the agent, and that in a passive sentence the subject does not always assume the patient role. This shows that semantic representations exist that differ greatly from its (deep) syntax (Emonds, 1991).

Circling back to the notion of proto-roles by Dowty, *Ursula* in 2.5 involves all four entailments of the Proto-Agent role as set out in 2.1, whereas *the pickaxe* in 2.6 would only involve the entailment of causality and movement, yet not of volition and sentience.

The pickaxe is then still a Proto-Agent, but to a varying degree. Nonetheless, *the ice* in 2.7 would still take on the Proto-Patient role. The implications that come with these different approaches on identifying semantic roles is further discussed in the next section that introduces PropBank.

The identification of thematic roles describes aspects of the syntax-semantics interface, specifically how the meaning of a verb sense supports certain grammatical relations and how that links to the semantic classification of its participants (Saeed, 2015). As such, where syntax is a device that helps to create grammatical and interpretable structures, semantics is the device that helps us derive meaning from those structures. The assumption of the current pipeline that prescribes a limited set of semantic roles to certain positions in the sentence runs the risk of generalizing syntactic structures over semantic representations.

Moreover, there is more to semantic roles than just the study of semantics. The question that semantic roles answer is that of *Who did what to whom?*, and choosing who to include in answering that question and who to omit can affect attitudes towards the participants in an event. For example, when immigrants are portrayed in the media as a threat, they are more likely to take on agentive roles, playing into the negative stereotype and feeding into hostile attitudes. However, when immigrants are portrayed as victims, they are more likely to take on the patient role, playing into the positive stereotype and feeding into sympathetic attitudes.

PropBank

To be able to computationally assign semantic roles to a given text, an annotated corpus that can be used as training data for machine learning approaches is exactly what is needed. This was the primary goal of the development of the Proposition Bank, or PropBank, to add predicate-argument structures on top of a treebank using semantic role labels for each verb argument (Kingsbury and Palmer, 2002). The labeling of the semantic roles is consistent, simple and functions for a general purpose, to support the training of automatic semantic role labelers Palmer et al. (2010). PropBank follows a 'per-predicate' annotation principle, meaning that each individual predicate in their own sense has its own 'frameset' (Bonial et al., 2012). A frameset consists of numbered argument labels, where ARG0 is the argument that shows entailments of the Proto-Agent, and ARG1 shows entailments for the Proto-Patient (Dowty, 1991). The higher arguments are not further generalizable, yet an effort has been made to consistently define roles across members of VerbNet classes (Palmer et al., 2010). Each argument is further defined in its frameset to contain more specific information. For example, the two participants that the verb hunting takes are not only labelled with ARG0 and ARG1, ARG0 is specified as *hunter* and ARG1 is specified as *hunted_entity*. Moreover, optional argument modifier roles are defined, such as location, cause, temporal, manner and direction to name a few (Bonial et al., 2012). Each verb sense, also in polysemous cases, comes with its own frameset, as well as syntactic variations in the possible expressions of that verb. As such, the Proto-Agent ARG0 roles are consistently labeled across syntactic variations, such as a change in voice. Even though the consistent labels of PropBank make it possible to generalize across datasets, it also makes it more challenging to estimate when two verbs's arguments have the same role and as such makes it more difficult to generalize across verb classes given its per-predicate roleset definitions.

The choice for using PropBank for adding semantic roles to the Microportrait

Pipeline is manifold. Most significantly are the readily available datasets on GitHub that are annotated with PropBank semantic roles. This access to a training, development and test dataset makes it easier and possible to train and develop a machine learning model for the semantic role labeling task at hand. Secondly, since the ARG0 and ARG1 roles correspond with the Proto-Agent and Proto-Patient roles as defined by Dowty, and given that these arguments make up 85% of arguments (Palmer et al., 2010), it makes a model trained on PropBank data highly effective, yet still aware for more semantic nuance.

Conclusion

This chapter discussed how stereotypes are theorized within social psychological literature and showed how stereotypical beliefs are constructed, maintained and shared through language. By discussing common approaches within the NLP research community, it provided reasons why the Microportrait Pipeline can help overcome some methodological challenges. By having an extractive pipeline that is transparent, stereotype-agnostic, genre-agnostic and does not rely on the subjectivity of annotators, it provides an approach to exploring the existence of stereotypes within natural language. Furthermore, this chapter also introduced some current shortcuts within the pipeline that can be further improved. The implementation and evaluation of these improvements will be further discussed in the next chapter.

Chapter 3

Methodology

Now comes the section that will discuss what has been done to find an answer to the research question. The main contribution of this thesis is to add semantic role labeling (SRL) to the Microportrait Pipeline. This has two main steps: 1) creating an SRL model for Dutch and 2) implementing this in the architecture of the pipeline. After these two steps are discussed, it will be explained how this change in the pipeline has been evaluated. Finally, the data that is used for this thesis will be introduced.

3.1 Microportrait Extraction Pipeline

This section presents the two main steps that this thesis has undertaken. First, it will further elaborate on the current heuristics of the dependency-based pipeline, which will then lead to how SRL has been implemented into the pipeline.

3.1.1 Semantic Role Heuristics

The main purpose of this thesis is to include an SRL system into the extraction pipeline, in order to create more detailed microportraits. As such, the current pipeline only uses the surface syntax of dependency relations to reason which semantic role an entity can take on. The three roles that are included in the current pipeline are `agent`, `undergoer` and `recipient`. The code now assigns an `agent` role when an entity holds the subject position related to the verb, and is not part of a copula construction/holds a predicative complement. A copula construction namely does not have an agentive relation, meaning the sentence holds information on a property rather than a role. Another rule is set in place that does not assign the `agent` role to the subject of a passive sentence, as this generally is not the `agent` of the verb, but the `undergoer`. Furthermore, an `undergoer` role is now ascribed when the head of an object relation falls under the PoS tags `verb` or `adjective`. Another check is implemented to account for passive structures, which checks whether the subject of the sentence is also the deep direct object. If this is true, it will assign the subject in a passive construction as the `undergoer` role. Even though this may be the case in quite some passive constructions, the subject in passive construction is not always the `undergoer`. The `recipient` role is now assigned when the dependency relation to its head is `obj2`, or in case of other prepositional phrases the entity is assigned a `prep-role`, where `prep` is substituted for the respective preposition.

3.1.2 Semantic Role Labeling

The heuristics of the original Microportrait Pipeline simplify how entities can play a semantic role in an event. As such, the SRL system that will be implemented into the pipeline is not a rule-based system reasoning over deep dependency structures, but a neural network that is able to capture complex linguistic patterns for accurate semantic role labeling. The reason for using a system trained for SRL, rather than a rule-based system, is to be able to capture more complex semantic representations than can be derived from its syntax, as discussed above. The model developed for this thesis has been trained on PropBank data, as introduced in Section 2.2.3. When a machine learning model is given enough data to train on, it can recognize a more diverse set of arguments, such as modifiers, negation, location, and manner. Furthermore, it can recognize patterns in which verbs can take on which alternations, knowing that *break* can take an instrument (or in PropBank labels ARG2-MNR) in a subject position. As such, the vastness of a machine learning model can learn more patterns in semantic representations than a rule-based system.

Figure 3.1 illustrates which steps have been taken to finetune each BERT model for SRL, the following sections will explain how these steps have been executed.

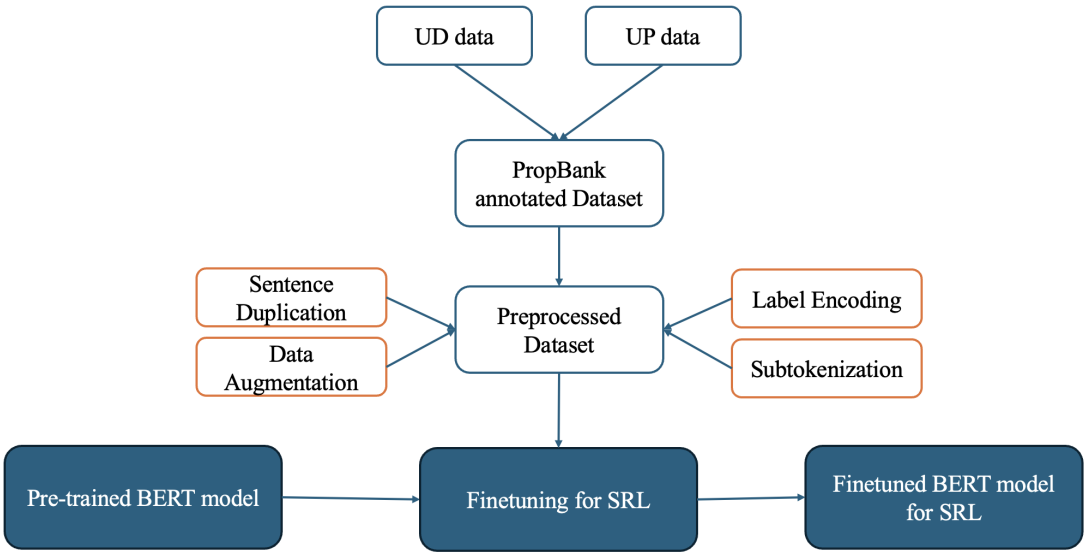


Figure 3.1: Finetuning BERT for SRL

XLM-RoBERTa or BERTje

The first step is to establish which pre-trained BERT model will be used, which is the blue box on the bottom-left. For the task of Dutch SRL, there are two models that compete for the best application: XLM-RoBERTa and BERTje. XLM-RoBERTa is a mix of an XLM model and the RoBERTa model. XLM models are language models that have cross-lingual pretraining objectives (Conneau and Lample, 2019), and RoBERTa is a BERT model pretrained on the CommonCrawl Corpus that consists of 100 languages (Conneau et al., 2020). This allows for an increase in pre-training dataset size, the possibility to exploit the cross-lingual pretraining objectives of an

XLM model, combined the same benefits of the unsupervised pre-training objectives of a BERT model; masked language modeling (MLM) and next sentence prediction (NSP) (Devlin et al., 2019). XLM-RoBERTa (XLM-R) is stated to outperform state-of-the-art monolingual models (Conneau et al., 2020), making it an efficient choice for a more complex downstream task like SRL.

BERTje is a monolingual Dutch model that is pretrained on a large and diverse dataset of 2.4 billion tokens and states to outperform the multilingual BERT (mBERT) model on regular downstream tasks - including SRL (De Vries et al., 2019). To be able to establish which model will work best for the task of SRL, three different scenarios have been tested of which the results can be found in Section 4.1:

1. A BERTje model has been finetuned on the LassySmall and Alpino dataset, which are both Dutch datasets with PropBank annotations.

2. An XLM-RoBERTa model has been finetuned on the LassySmall and Alpino dataset, feeding it only Dutch data during finetuning.

3. An XLM-RoBERTa model has been finetuned on the LassySmall, Alpino and the majority of the English UP2.0 dataset, in order to exploit its cross-lingual pre-training objectives during the finetuning stage.

The BERTje model has not been finetuned with both English and Dutch data, since its pretraining was on entirely Dutch data. As a result, there are no cross-lingual patterns the model can pick up from adding English data to its finetuning process. Furthermore, only a majority of the English data was used for finetuning, as the model would otherwise get too big for the available computational units.

Task Definition

Next up is the step of finetuning the model for SRL, which is the middle blue box in Figure 3.1. XLM-R and BERTje are finetuned for the task of SRL using the Universal Propositions (UP) data.[1] The datasets of Alpino and LassySmall are used to finetune both models. The UP data is readily available on GitHub with only predicate and argument information and still needs to be mapped onto its Universal Dependency (UD) counterpart to get lexical information. This is done by using the script on the UP github page.[2] This results in a PropBank annotated dataset with SRL annotations for Dutch in the CoNLL-U format. The dataset used is from UP2.0, which differs from the UP1.0 by having higher quality SRL annotations for English, which translates directly onto the quality of the projected annotations on the target language (Jindal et al., 2022). Moreover, UP2.0 is trained to be able to predict both span- and dependency-based semantic roles, where only the head of a phrase is assigned a label (Jindal et al., 2022). Nonetheless, this thesis will only train for argument head identification and classification. This decision is made based on three reasons. First, one pre-trainig objective of XLM-RoBERTa is masked language modeling, which masks a token and has the model predict the masked token (Devlin et al., 2019). Other BERT architectures, such as SpanBERT (Joshi et al., 2020), are pretrained with the objective to recognize spans through span-level masking. As a result, XLM-R is much better suited for token classification, rather than span classification. The second reason is that the head of a

[1] https://github.com/UniversalPropositions
[2] https://github.com/UniversalPropositions/tools/tree/main

phrase exists of one token, meaning that there is either a match or not. With span-level prediction, partial-match predictions should be taken into account when evaluating the performance. This makes head-prediction more efficient in establishing which model is the best to use for Dutch SRL. The third reason is that the Microportrait Pipeline picks spans apart to gather the descriptions. Where modifiers are often part of a phrase, as in *the pious Muslim smiled*, *pious* becomes a property of the label *Muslim*, which makes it reasonable for the role of ARG0 being assigned to only *Muslim* and not the whole phrase *the pious Muslim*. As such, given the application of the model as part of the Microportrait Pipeline, argument head identification and classification suffices for the purpose.

Preprocessing

Now that we have a dataset with PropBank annotations, it needs to undergo a few preprocessing steps to ensure that the BERT models understand what they are being fed. This moves us further down the line of finetuning as shown in Figure 3.1. First of all, this is done by duplicating each sentence for each predicate, to ensure that each sentence is a training instance for only one predicate. Each sentence is then augmented to inform XLM-R whether it is dealing with a predicate, or a potential argument. The augmentation approach is based on the paper of Khandelwal and Sawant (2020), where a special token `[NO]` or `[PRED]` is prefixed to each token that indicates whether it is a predicate or not. This results in tokens of the form:

1. *[NO]I [PRED]walk [NO]home [NO]tonight*

The tokens are further subtokenized using Sentence Piece, which is a language-independent subword tokenizer and is trained directly from raw sentences (Kudo and Richardson, 2018). These characteristics of Sentence Piece make it an end-to-end language-independent system, which is useful when applying it on different languages at the same time. Subtokenizing is necessary for BERT models as this avoids the issue of defining a word and eliminates out-of-vocabulary words. As such, the subword tokenizer has a vocabulary of around 30,000 tokens, and if a token does not exist in that vocabulary, it is split up into subtokens so it aligns with the known vocabulary. This can result in sentence 1 above being presented as shown in 2. Each subtoken is separated with a comma in this example:

2. *[NO], I, [PRED], walk, [NO], home, [NO], to, ##night*

This shows how *tonight* is split up into two words, *to* and *##night* to fit the known vocabulary. Lastly, the semantic role labels should be encoded for the BERT models, as it cannot understand strings but only integers. As such, a simple dictionary is made mapping each label of SRL (A0, A1, A2, A3, etc.) to a number between 0 and 55, the total amount of unique labels, including all argument modifiers.

This results in the preprocessed dataset as shown in Figure 3.1. The preprocessed data has been fed to XLM-RoBERTa and BERTje, which eventually results in a finetuned BERT model for the task of SRL.

Evaluation

The final step for establishing whether the finetuned model is of any quality is by performing an evaluation. The models have been evaluated using the Dutch UP test

data set. The test dataset exists of only Dutch data, so no English data is used for evaluation, as English SRL is not the training objective. The evaluation is done using standard precision, recall and f1 metrics. In Section 4.1, the results of each model will be discussed.

Pipeline Implementation

After the best model has been selected, SRL still needs to be implemented into the Microportrait Pipeline. To do this, the following steps have been taken as shown in Figure 3.2.

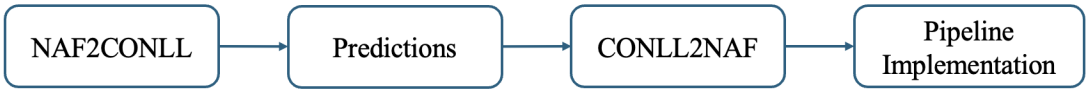


Figure 3.2: Pipeline Implementation of SRL

The Microportrait Pipeline is designed to retrieve information from NAF files. NAF stands for the NLP Annotation Format, which can hold information about a text on different linguistic levels, such as morphosyntactic information, dependency information and SRL information. Each level has its own layer in the file, following an XML layout (Fokkens et al., 2014). The first step is then to convert each NAF file to a CoNLL format. The NAF files have been converted into .txt files of the CoNLL format. This keeps the information of each term id as set out in the NAF file, together with token, lemma and predicate information in each column. The CoNLL files are of the following format, where the prediction column is only added after the SRL model has made its predictions:

term_id	**token**	**lemma**	**predicate**	**prediction**
t_1	Hier	hier	_	_
t_2	zitten	zitten	V	_
t_3	ongetwijfeld	ongetwijfeld	_	_
...				

Table 3.1: CoNLL-style tsv format for SRL predictions

Predicate identification is done by a rule during conversion that assigns a predicate role to a verb. The part of speech information is directly extracted from the NAF file itself, which includes PoS information from a morphosyntactic parser based on Alpino. This does come with the limitation that non-verbal predicates are omitted. However, the Microportrait Pipeline assigns non-verbal predicates that exist in copula constructions as a property within the extracted portraits, not a semantic role. Adjectival predicates in a sentence such as *the woman is ambitious*, makes *ambitious* a property of the label *woman*, and does not assign *woman* a semantic role ARG1 for the adjectival predicate *ambitious*. Given that nominal predicates are underrepresented in any natural language dataset (Orlando et al., 2023), these non-verbal predicates are not further accounted for in this application of predicate identification.

The finetuned XLM-RoBERTa model is then used to make predictions on the file, which is the second box in Figure 3.2. The predictions are stored in the final column in

the CoNLL formatted file. Once the files with SRL predictions exist, the final columns can be converted to a new SRL layer in each respective NAF file, as indicated in box three of Figure 3.2. This is done by using pre-defined classes from the KafNafParser module.[3] By creating a separate class for each predicate, and adding semantic roles to its corresponding predicate, this predicate class with semantic roles can then be directly added to the parsed NAF object. With the new SRL layer, the new NAF object is dumped into a new folder of NAF files that includes semantic information.

The final step of Figure 3.2 is to adjust the pipeline. The pipeline has then been adapted to retrieve SRL information from the NAF files directly, instead of following a set of heuristics that is based on dependency structures. Now the Microportrait Pipeline produces so-called nanoportraits, which are portraits for each entity within one sentence boundary. This shows which labels, properties and roles exist within the same portrait for one entity mention. This means that the evaluation takes place on a portrait-level, rather than for each extracted description. Do note that the final step of coreference resolution is not yet implemented for this iteration of the pipeline, meaning that the microportraits this pipeline produces are still sentence-level nanoportraits.

3.2 Evaluation

To properly evaluate the two pipelines, let's recall that this thesis is interested in finding out to what extent the information the pipeline yields changes with the addition of SRL. The research question is: *to what extent can the inclusion of semantic role labeling in the microportrait extraction pipeline increase the granularity and improve the accuracy of microportrait extraction in Dutch News?* From this we want to know what the differences are between the patterns that the syntax-based pipeline produces and the patterns that the SRL-based pipeline produces. As such, the goal is to compare those patterns to find how the granularity changes, and what effects that has on the information the pipeline presents. In other words, we want to compare if and how the SRL predictions improve the information the pipeline produces.

The goal then is to compare the results of the updated pipeline, to that of the original pipeline. Both the original pipeline and the updated SRL pipeline are applied to the data, which results in nanoportraits that are still based on dependency-heuristics and nanoportraits that contain SRL information. The performance of the SRL system is already evaluated, as can be seen in Table 4.1 and A1. As such, no attempt will be made to evaluate the performance of the SRL system on the microportrait data. The evaluation will further take place on two levels. First, patterns will be established on a quantitative level, and then further explored on a qualitative level.

3.2.1 Quantitative level

The quantitative level will evaluate the difference between the two pipelines from a bottom-up approach. As described in Fokkens et al. (2018), this is done by calculating the Pointwise-Mutual Information scores of descriptions that co-occur for the same microportrait or group of interest. PMI calculates how often two terms of interest co-occur in a given dataset compared to how often they would appear by chance. This comparison is based on the intuition that it indicates to what extent those two terms are associated with each other (Church and Hanks, 1990). As a measure of association,

[3] https://github.com/cltl/KafNafParserPy/tree/master

these outcomes show which descriptions are more typically used for the group of interest compared to another group, comparing the attitudes towards minority and majority groups. The formula behind PMI between a target word w and a context word c is:

$$PMI(w, c) = log_2 \frac{P(w, c)}{P(w)P(c)} \tag{3.1}$$

This then illustrates which label, property or role is more associated with one social group compared to another. The PMI scores are calculated in Python following the equation above. To calculate the PMI scores, the tokens are lemmatized, made lowercase, and punctuation and stop words are filtered out using the Spacy library. Furthermore, before the PMI scores are calculated, the co-occurrences are filtered with a minimum frequency count of 5. That is because PMI can become unreliable when the co-occurrence counts become very low, creating a bias towards low-frequency word pairs (Church and Hanks, 1990). As such, this threshold follows Church and Hanks (1990) to avoid word pairs that occur less than 5 times to dominate the PMI-scores.

The PMI scores are calculated for both the output of the Microportrait Pipeline based on syntax as well as on SRL. This results in a dataset of 4393 files for the SRL output, and 4459 files for the syntax-based output. The reason that some files do not have any SRL output is because no predicates were identified in the files, meaning that no semantic role labeling could take place. The datasets are nonetheless reduced to contain the same 1956 files for both outputs, since the full dataset takes up quite some computing effort. By calculating the PMI scores directly on the microportrait output, it is possible to establish which descriptions are most associated with which entity. This provides a more informative output on how the portraits change between the two pipelines compared to using a window on the raw text itself.

To discover differences in patterns, the two groups originally up for interest in Fokkens et al. (2018) are reused for this evaluation, however, the target words are selected specifically for this thesis and are not taken from the original study. The target words w for the minority group are chosen to be *Islam* and *Moslim*, and the target words w for the majority group are *Nederlanders* and *kerk*. Each target word is a label that occurs within its own portrait, with the properties and roles that are assigned to the query label. Each portrait that contains the target label is extracted, and then the PMI scores are calculated for the descriptions to discover which descriptions are most associated with a certain label. It does occur that labels remain part of the descriptions, as one portrait can contain multiple labels. This happens when there are multi-word expressions that are not named entities, determiners for the entity, as well as in case of appositional relations. However, this does not always have to be a relevant label that further describes the entity of a nanoportait. This phenomenon is also discussed in Section 4.2.1, as well as in Section 5.2.1. For each target word w, the 10 descriptions c in the form of labels, properties and/or roles within a portrait with the highest PMI scores are used for comparison. This measure obtains the most typical descriptions for a word of interest that is linked to a target group. By comparing the outcome of this measure for both pipelines, it can show differences in patterns that occur for extracting descriptions, most likely in roles and behaviors.

3.2.2 Qualitative level

Manual Annotation

At the qualitative level, the evaluation will also look at differences in patterns, focusing more on the extracted microportraits and their descriptions. For this purpose, 1 article is annotated manually to set a gold standard for the comparison. The annotation is done by the author, about whom further details can be found in the data statement in Appendix 6. By doing a manual annotation there is a semi-independent standard for the gold labels, as they are annotated before seeing any output of either pipeline. This is to prevent the annotations to be influenced and show bias towards the SRL system that this thesis created, compared to the syntax-based pipeline it tried to improve. The annotations are *semi*-independent because they are only annotated by one annotator, meaning that an inter-annotator-agreement cannot be calculated. The articles are selected when it mentions a target group, following Fokkens et al. (2018), this means the Dutch words for Muslim, Christian, Jewish, Belgian, German, Moroccan, Turkish, Dutch and derived terms. The labels, properties and roles for each group are annotated, where the roles are labeled following the PropBank annotation guidelines (Bonial et al., 2012).

The format of the annotation follows that of a nanoportrait, which is a sentence level microportrait. The guidelines are built upon the set-up as set out in Fokkens et al. (2018):

1. Within a sentence, mark all labels used to refer to an entity, each entity within a sentence gets a portrait.

2. Mark all the properties of the label and give it the same portrait identifier as the label.

3. Identify the argument the entity takes on for the predicate in the sentence, if applicable. In case of doubt, refer to the PropBank framefiles.

Portrait Identifier	**Relation**	**Description**
ID1	label	Islam
ID1	property	radical
ID2	label	Muslim
ID2	A0	commit

Table 3.2: Illustration of different nanoportraits

Table 3.2 illustrates what the annotated nanoportraits would look like. All entities within a sentence are extracted by their label, and its corresponding properties and roles are added to the same portrait as indicated by the identifier. As such, the table shows how the phrase *radical Islam* is represented in portrait ID1 with the label *Islam* and the property *radical*. ID2 shows part of the sentence *that wasn't committed by Muslims*, where *Muslim* is the label, and this entity takes on the ARG0 role for the verb *commit*. These examples are taken directly from the data at random.

Comparative Analysis

Once the manual annotations are finalized, as well as the output from both pipelines, their results can be compared for their quality. The difference between the two pipelines should become clear when we want to answer the question *who did what to whom?* This question captures the semantic representation of an event and its participants. To further evaluate how the answer to this question changes, the following guiding questions are used to guide the qualitative analysis:

1. Are there any instances where the SRL predicted label is more accurate than the heuristics label?

2. Are there instances where the SRL predicted label provides more information of a higher granularity than the heuristics label?

By evaluating the outputs of the two pipelines with these guiding questions, it will illustrate if and how SRL improves the Microportrait Pipeline. This evaluation will be done manually to properly assess the content of the semantic arguments. The assumption is that the SRL predictions are more robust for syntactic variations, and can provide more information on which role the prepositional phrases play in an event.

3.3 Data

The data that is used for this thesis is the same dataset that was used in the original publication by Fokkens et al. (2018). This dataset consists of news articles from several Dutch news outlets, of which an overview can be found in Table 3.3. The articles were selected between September 5 2016 and March 15 2017, yet since they are only used to evaluate the addition of SRL, this should pose no issue.

National Newspapers	**Online Newssites**	**Blogs**
Algemeen Dagblad	AD.nl	Joop.nl
De Telegraaf	Telegraaf.nl	Geenstijl.nl
NRC Handelsblad	NRC.nl	
Trouw	Trouw.nl	
De Volkskrant	Volkskrant.nl	

Table 3.3: Overview of Media Outlets

The data consists of NAF files that contain several layers of information. It is tokenized, lemmatized and syntactically parsed. This thesis adds the SRL layer to each NAF file. Each article that is part of the dataset is of political nature, meaning that they are selected based on the mentioning of any political actors within the Netherlands. A full data statement can be found in Appendix 6.

Chapter 4

Results

This section will present the results of the evaluation of the old and the new pipeline. The implementation of SRL in the pipeline has been discussed in the previous chapter and the evaluation strategy has been introduced. This chapter showcases the results, where the results of the SRL model will be discussed in Section 4.1, the quantitative evaluation in Section 4.2.1 and the qualitative evaluation in Section 4.2.2. The outcome of this section will be further discussed and interpreted in the discussion in Chapter 5.

4.1 BERTje versus XLM-RoBERTa

As there are no off-the-shelf SRL systems for Dutch, this thesis created its own SRL model by testing different finetuning approaches. The three approaches for the SRL system were 1) feeding XLM-R only Dutch data, 2) feeding XLM-R both Dutch and English data and 3) feeding BERTje Dutch data. This section will report the results of each finetuning approach, and establish how the decision for XLM-RoBERTa has been made.

	BERTje			**XLM-R Dutch**			**XLM-R Dutch & English**		
	P	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
A0	0.86	0.86	0.86	0.82	0.84	0.83	0.86	0.87	**0.87**
A1	0.79	0.85	0.82	0.79	0.84	0.81	0.80	0.87	**0.83**
A2	0.68	0.74	0.71	0.66	0.74	0.70	0.70	0.77	**0.74**
AM-MNR	0.64	0.82	0.72	0.61	0.76	0.67	0.68	0.85	**0.76**
AM-MOD	0.87	0.94	0.91	0.85	0.94	0.89	0.93	0.97	**0.95**

Table 4.1: Comparison of precision (P), recall (R), and F1 scores for the three models.

Table 4.1 shows the performance of the three models for the most prominent classes for this SRL task, which are also the classes that are most useful for extracting microportraits. Reporting the metrics for the system overall would show much lower results than we are looking for, since for the purpose of the Microportrait Pipeline, the reported classes are the most important. The best results can be found for the XLM-R model that has been finetuned on both Dutch and English data. For the Proto-Agent and Proto-Patient roles, the f1 score is respectively 0.87 and 0.83. Additionally, the performance for `AM-MNR` and `AM-MOD` are also high, which makes them also relevant la-

bels to include in the microportrait extraction pipeline. Nonetheless, each label that is predicted and extracted by the pipeline will be included, yet for the lesser performing classes, it is necessary to be critical of the predicted labels. The results of the model can be found in Table 4.1. The full results can be found in the appendix in Table 1. The results show that the XLM-R model trained on both Dutch and English data slightly outperforms the Dutch monolingual model BERTje. Given the importance of a high-perfomance SRL model for the Microportrait Pipeline, this thesis used the XLM-RoBERTa model that has been finetuned on both English and Dutch data as its SRL system.

4.2 Microportraits Evaluation

This section presents and discusses the results of the evaluation. First, the PMI scores that indicate what the most associated terms are per keyword for each pipeline are compared to each other. Secondly, the portraits for each pipeline are further compared and analyzed on a qualitative level. Patterns and errors are discussed and explained.

4.2.1 Quantitative Level

To calculate the PMI-scores, four keywords are selected to show different outcomes for two different social groups. The keywords are the labels about which properties and roles are extracted. For Muslims, these are the words *Moslim* (Muslim) and *Islam*, and for Dutch people, these are *Nederlander* (Dutchman) and *kerk* (church). The PMI scores will then indicate which properties and roles are most associated with the label *Moslim* or *Nederlander* within the subset of 1956 files as explained in Section 3.2.1.

SRL				**Syntax**			
Relation	**Descriptor**	**Joint Freq**	**PMI**	**Relation**	**Descriptor**	**Joint Freq**	**PMI**
label	democraten	6	7.974	property	anti-migratie	5	7.718
property	onverzoenlijk	5	7.881	property	anti-	5	7.455
property	dominaat	5	7.711	property	sji'itisch	6	7.303
property	will	13	7.259	label	democraten	6	7.133
property	omgangsvorm	8	6.974	property	anti-establishment	5	6.718
property	in_tolerant	5	6.711	agent	wereld_toneel	6	6.718
property	dominant	9	6.407	property	dominaat	5	6.580
property	jihadisme	6	6.336	property	tweederangs_religie	8	6.548
property	radicaal	75	6.245	property	onverzoenlijk	5	6.232
A1	aan_praten	8	6.237	property	will	13	6.209

Table 4.2: PMI scores for w = *Islam* including all descriptions.

Table 4.2 shows the PMI scores of the SRL pipeline on the left and that of the syntax pipeline on the right. For each pipeline, the relation is presented stating whether the most associated term is a label, property or role, the descriptor column shows the lemmatized token, the joint frequency shows how often the target word w and the context words c co-occur within a portrait, and in the most right column the PMI score is presented from most to least associated for the 10 highest PMI scores.

What the results show in Table 4.2 is that there are both similarities and differences between the portraits that the SRL pipeline and the syntax pipeline extract. Even though the type of relations show a similar distribution, there is quite some difference in the descriptions that are the most associated. For example, the SRL pipeline has among its most associated descriptions with *Islam* the properties *Jihadism* and *radicaal* (radical), whereas the syntax pipeline does not have these two terms in its top 10, but rather on place 44 and 19 respectively. Additionally, both *Jihadism* and *radical* have higher joint frequency counts for the syntax pipeline then for the SRL pipeline. Nonetheless, they still have a lower PMI score, even though a higher nominator would result in a higher PMI score. This can be explained by the relation that a context word c with a lower joint-frequency count can have a higher PMI score because it is then more specific to the query, as a result of the absolute frequency of c or w being lower. If the overall count of either target word and context word is lower, this would result in a stronger association and higher PMI score.

Additionally, the higher joint frequency counts for the syntax pipeline compared to the SRL pipeline can be attributed to its syntax-based extraction. Meaning that every time the terms are syntactically related, they are extracted. This observation is also underlined by the inclusion of *wereld_toneel* (world stage), *anti-* and *anti-migratie (anti-migration)* in the top 10 most associated terms for the syntax pipeline, despite it not being extracted by the SRL pipeline at all. This illustrates that the syntax pipeline extracts many more descriptions, of which some appear to be noise, such as the `agent` relation for *wereld_toneel*. On the other hand, such descriptions are not included in the portraits extracted by the SRL pipeline. This can either result in more selective portraits that are better generalized, or it can result in the ommission of important descriptions.

SRL				**Syntax**			
Relation	**Descriptor**	**Joint Freq**	**PMI**	**Relation**	**Descriptor**	**Joint Freq**	**PMI**
property	zoenen	6	8,092	property	baarddragen	8	7,144
property	baarddragen	8	8,092	property	zoenen	6	6,559
property	staat_discriminatie	6	7,677	undergoer	vernederlandsen	5	6,296
A0	vernederlandsen	5	7,607	property	staat_discriminatie	6	6,269
property	en/of	8	7,507	label	900.000	8	5,286
label	900.000	8	7,392	in-rol	900.000	8	5,286
property	bidden	31	7,139	property	schiet_partij	6	5,205
property	radicaal	160	6,871	property	en/of	8	5,189
property	schiet_partij	6	6,870	property	bidden	31	5,087
property	extremistisch	20	6,829	property	extremistisch	23	5,082

Table 4.3: PMI scores for $w = Moslim$ including all descriptions.

For the target word $w = Moslim$ in Table 4.3 we see that similar observations can be made as for Table 4.2. Which are 1) the syntax pipeline extracts many more descriptions that the SRL pipeline does not extract, 2) higher PMI scores for the same descriptors with the same joint-frequency counts if extracted by the SRL pipeline, 3) higher joint-frequency counts for the same descriptor if extracted by the syntax pipeline. The first observation can only be supported with the top-10 most associated terms by the description *900.000* that also occurs as an in-rol. Nonetheless, the total amount

of descriptions for the syntax pipeline are 623 compared to 191 for the SRL pipeline, which shows that the SRL pipeline extract significantly less descriptions. The second observation is supported by the descriptors *baarddragen* (beard-wearing), *bidden* (pray) and *schiet_partij* (shooting), to name a few. Each of those terms holds the same joint frequency, yet the PMI scores of the SRL pipeline are higher. This would indicate that the total amount of descriptions extracted for the SRL pipeline is again lower than that of the syntax pipeline, which would result in higher associations. The third observation is supported since *radical* does occur in the top 10 for the SRL pipeline, whereas it only occurs at place 14 for the syntax-scores, with 170 co-occurrence counts compared to 160 for the SRL pipeline. Given the higher joint frequencies, this further indicates that the syntax pipeline may extract every occurrence of the word when it is syntactically related, instead of filtering for semantic associations. Furthermore, this can come with the implication that some descriptions which are highly associated with the query term are not represented as such by the syntax-pipeline.

An interesting observation from these scores is that *vernederlandsen* (roughly translated to Dutchify) is most associated in the A0 role for the SRL pipeline, and as `undergoer` for the syntax pipeline. *Vernederlandsen* occurs in the short sentence: *Moslims vernederlandsen*, which can be seen as semantically ambiguous. First of all, the dependency parser that the syntax pipeline bases its semantic roles on, assigned *Moslims* the `direct object` role, which results in an `undergoer` label. It is possible to state that Muslims undergo the process of becoming more Dutch, as it can be something that just happens to someone living in the Netherlands. However, just as the process of integrating within a society, there is an agentive aspect to the one integrating. This would mean that *Moslims* are indeed the agent of the verb, as predicted by the SRL system. Nonetheless, depending on the context, the connotation of the verb *vernederlandsen* changes, which can in turn change how a semantic role would be assigned.

Another observation in Table 4.3 is that *900.000* appears in both pipelines as `label`, and for the syntax pipeline as an `in-rol` as well. As such, *900.000* appears in the articles as a determiner for the noun *Moslims*, in the full phrase *circa 900.000 moslims in Nederland* (around 900.000 muslims in the Netherlands). *900.000* then becomes another label that describes Moslims, since it holds the determiner relationship to *Moslims*. However, it is not clear why *900.000* appears again in an `in-rol`, as it is not related to *Moslims* via a preposition. As such, if we were to remove the labels from the most associated terms to avoid the noise it brings, we would get the following representation as shown in Table 4.4.

SRL				Syntax			
Relation	**Descriptor**	**Joint Freq**	**PMI**	**Relation**	**Descriptor**	**Joint Freq**	**PMI**
A0	vernederlandsen	5	8.282	undergoer	vernederlandsen	5	7.919
property	bidden	17	7.022	agent	slachtoffer_rol	6	6.398
property	radicaal	102	6.976	property	bidden	17	5.877
property	extremistisch	12	6.767	property	radicaal	102	5.817
A0	vermoorden	53	6.205	property	extremistisch	12	5.767
property	circa	6	6.182	door-rol	vermoorden	55	5.651
property	gematigd	8	6.182	property	gematigd	8	5.123
property	2004	6	5.624	undergoer	haten	9	5.067
A0	op_houid	6	5.143	van-rol	heilig	10	4.909
A1	haten	7	5.083	door-rol	helaas	12	4.901

Table 4.4: PMI scores for $w = Moslim$ without any labels for descriptions.

By removing the extra labels in the portraits, we can see how different descriptions rise above, especially more semantic roles. Most notably is the observation regarding the relation that is assigned for the verb *vermoorden* (murder). The SRL pipeline assigns *Moslims* the A0 role for the verb *to murder*, whereas the syntax pipeline assigns the `door-rol` (by-role). Despite the possibility to draw the conclusion that the `door-rol` is assigned in a passive construction, the syntax pipeline fails to assign the agent role to the entity. This shows the difference in the SRL pipeline finding an answer to the *who did what to whom?* question, and predicting that the entity described in the `door-rol` is the one who did something, whereas the syntax pipeline represents this as a syntactic relation, which, to an extent, answers the question *which syntactic relation does the entity have to the verb?*.

SRL				Syntax			
Relation	**Descriptor**	**Joint Freq**	**PMI**	**Relation**	**Descriptor**	**Joint Freq**	**PMI**
label	02:45	7	7,723	label	02:45	7	6,778
property	rechtgeaard	9	7,723	property	anti-marokkanen	6	6,363
property	akp_steunen	6	7,723	property	marrokkaans	8	6,078
property	aanvoering	14	7,530	property	rechtgeaard	9	6,041
property	marrokkaans	8	7,023	property	aanvoering	14	5,885
property	kwellen	6	6,986	property	kwellen	6	5,778
property	stemgerechtigden	10	6,723	label	dagdroom	5	5,515
property	hevig	6	6,723	in-rol	grossier	7	5,498
property	surinaams	79	6,354	agent	rechtgeaard	6	5,456
property	hardwerken	18	6,338	van-rol	2.832	8	5,456

Table 4.5: PMI scores for $w = Nederlander$ including all descriptions.

Moving on to the target word $w = Nederlander$ in Table 4.5, we can recognize similar patterns as the previous to target words. That is, SRL extracts significantly less descriptions, PMI scores are higher for the SRL pipeline, and joint-frequency counts for the same descriptor are higher for the syntax pipeline. That is, *hardwerken* (hard-working) does occur as highly associated for the SRL portraits, yet only occurs in 21st

position for the syntax portraits with 20 co-occurrences compared to only 18. Additionally, the descriptions of *anti-marokkanen* (anti-moroccan), *dagdroom* (daydream) and *grossier* (wholesaler) do not occur at all in most associated descriptions of the SRL pipeline. These differences in descriptors show how those most associated with the term *Nederlander* for the SRL pipeline are more descriptive of the query term *Nederlander*, such as *marrokkaans* (Moroccan), *stemgerechtigden* (eligible voters), *surinaams* (surinam) and *hardwerken* (hardworking). The descriptions as extracted by the syntax pipeline on the other hand, such as *dagdroom*, *grossier* and *anti-marrokkaans*, do not have such direct descriptive characteristics of the target word. Since *grossier* is also an `in-rol`, it appears that the syntax pipeline includes more syntactic associations in its portraits than extracting semantic relations. This then illustrates how the differences in extracted descriptions can change when either syntax or SRL is applied. Both pipelines produce text representations based on either semantic or syntactic information, which directly reflects in how the descriptions are extracted and which descriptions are extracted.

Additionally, what Table 4.5 shows is how labels such as *02:45* are still included in the portrait, without adding much information, similar to Table 4.3. In the articles, *Nederlander* holds an appositional relation to the time indication *02:45*. This results in *02:45* becoming a part of the portrait as another label to refer to a Dutch person, despite this not being the case. As such, what would happen if these labels were excluded from the PMI calculation, to avoid the chance of having nonsensical labels create noise in the results? This is shown for the target word *Nederlander* in Table 4.6.

SRL				**Syntax**			
Relation	**Descriptor**	**Joint Freq**	**PMI**	**Relation**	**Descriptor**	**Joint Freq**	**PMI**
property	marrokkaans	8	7,637	property	marrokkaans	8	7,637
property	aanvoering	7	7,145	undergoer	wijzijnnederland.nu	7	7,337
property	surinaams	60	6,622	undergoer	vuijs	5	6,489
property	autochtoon	24	6,337	property	aanvoering	7	6,444
property	antilliaans	13	6,257	om-rol	stateloos	9	6,378
property	hardwerken	10	6,136	agent	ex-pvda	7	5,719
property	manifest	7	5,823	property	surinaams	60	5,671
A2	terug_geven	9	5,463	voor-rol	nachtmerrie	5	5,659
property	marokkaans	117	5,443	undergoer	motto	7	5,621
property	migratie_achtergrond	14	5,299	om-rol	bepleiten	10	5,233

Table 4.6: PMI scores for $w =$ *Nederlander* without any labels for descriptions.

What this shows is the difference in most associated properties and roles with the term *Nederlander*. It shows a stark contrast in different ways of describing types of *Nederlander* between the two pipelines, where the SRL pipeline shows different ways of describing a *Nederlander*, the syntax pipeline has more variation and noise in its associations. That is, the associations for the syntax pipeline include semantic roles for nouns, such as *motto*, adjectives like *ex-pvda* and website handles *wijzijnnederland.nu* (we are the Netherlands). When leaving out the labels from the portraits to find associations, it shows that the SRL pipeline creates an overview of different types of *Nederlander*, which is illustrated in figure 4.1

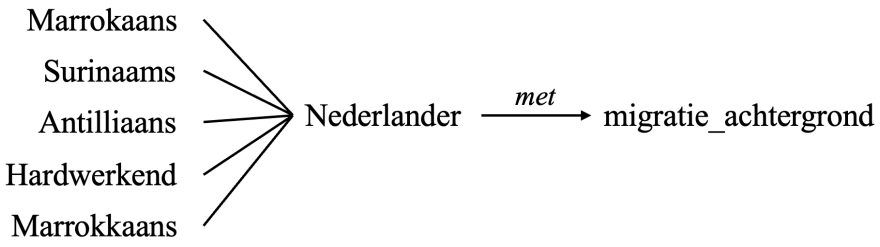


Figure 4.1: Different ways of describing *Nederlander*

When attempting to create a similar graph for the most associated descriptions for *Nederlander* in the syntax pipeline, it becomes more difficult to establish how each semantic role and prepositional role connect to the target word. Besides *Marrokkaans* and *Surinaams*, the noise that is included in the syntax-pipeline representation becomes more apparent than the information that is desired.

SRL				**Syntax**			
Relation	**Descriptor**	**Joint Freq**	**PMI**	**Relation**	**Descriptor**	**Joint Freq**	**PMI**
property	hervormd	5	9,965	property	outreach	8	8,335
property	roomskatholiek	8	9,158	property	hervormd	6	8,183
property	troon_rede	6	8,643	agent	sjoel	6	8,183
label	scheiding	17	8,498	property	victory	8	8,013
property	protestants	25	8,101	agent	homohuwelijk_jubileum	8	7,598
label	protestants	19	7,705	agent	europride	8	7,598
property	reformeren	5	6,672	property	roomskatholiek	9	7,335
property	katholiek	7	5,807	agent	regenboog_vlag	8	7,220
property	verhouding	5	5,763	undergoer	europride	6	7,183
property	feest	6	5,743	undergoer	homohuwelijk_jubileum	6	7,183

Table 4.7: PMI scores for $w = Kerk$ including all descriptions.

Finally, Table 4.7 also shows the same patterns as the other three tables but now for the target word $w = kerk$. The SRL pipeline extracts less descriptions, the PMI scores are higher for the SRL-scores and joint-frequency counts for the syntax pipeline are higher than the SRL pipeline. Furthermore, the syntax pipeline also extracts more roles, such as `agent` and `undergoer`. What can again be noted just as in Table 4.6, is that for this role extraction those roles are associated with nouns, which makes the church both an `agent` and `undergoer` of *europride* and the *homohuwelijk_jubileum* (gay marriage anniversary). This would indicate that there is some error in how the syntax pipeline assigns roles within a portrait, which in turn results in more noisy representations.

Moreover, many of the most associated terms with the term *kerk* in the syntax-portraits do not occur at all in the most associated terms of the SRL pipeline, such as *regenboog_vlag* (rainbow flag), *homohuwelijk_jubileum* and *sjoel* (Dutch shuffleboarding). This observation was also made for Table 4.3. To find out why these descriptions are included in the most associated terms with their respective relation, this has been further inspected for the term *homohuwelijk_jubileum*. What this shows is that this

term appears in the following sentence:

(1) Sommige kerken hijsen de regenboogvlag ter ere van het
some churches hoist the rainbow.flag in honour of the
homohuwelijkjubileum.
gay.marriage anniversary
'Some churches hoist the rainbow flag in honour of the gay marriage anniversary.'

This shows that the church does do something to the *rainbowflag* and is involved with the *gay marriage anniversary*. As such, the SRL pipeline did not include these terms in its representation at all, whereas the syntax pipeline did include them, yet in the wrong role. Nonetheless, extracting the prepositional roles does not include functional idioms such as the idiom *ter ere van* (in honor of) to narrow down in what role a term occurs. What this shows is that the SRL pipeline perhaps generalizes too much when assigning roles as it loses information that exists within a sentence, whereas the syntax pipeline does include this information, yet assigns the wrong roles which then makes the representation more noisy. This again underlines the differences in representation when choosing the SRL pipeline versus the syntax pipeline.

To further investigate why some of the patterns that have been discussed in this section occur, Table 4.8 shows some aggregated counts for each of the PMI calculations per target word.

	SRL			**Syntax**		
	Total descriptions	**Avg/ portrait**	**Median PMI**	**Total descriptions**	**Avg/ portrait**	**Median PMI**
Islam	156	x	2,804	487	x	0,882
Moslim	191		2,448	623		0,208
Nederlander	268		1,865	873		-0,181
Kerk	41		4,168	193		2,137

Table 4.8: Metadata PMI scores

This information supplements the previous patterns by underlining that the extracted information in the syntax pipeline is significantly more when it comes to quantity then the SRL pipeline. The total descriptions column shows how many descriptions are used to calculate the PMI scores, which shows that the syntax pipeline produces more than three times the amount of descriptions. This is a result of the pipeline extracting descriptions that are syntactically close to the respective portrait, even though they may not be as informative, such as *sjoel* (Dutch shuffleboard) in Table 4.7. Moreover, by having so many extracted descriptions, this significantly inflates $P(c)$ in the denominator of the PMI formula, which explains why the PMI scores for the syntax pipeline are structurally lower.

Furthermore, the average amount of descriptions per nanoportrait is double the amount for the syntax pipeline. This can explain the more dense semantic representations being extracted with the SRL pipeline, instead of all syntactic associations in the syntax pipeline. The portraits from the SRL pipeline are then less noisy and capture the semantics of a target word or entity better. However, this also underlines that this

can mean the amount of information the pipeline extracts decreases, as it does miss out on quite some descriptions. This will be further analyzed on the qualitative level.

The last column in Table 4.8 shows the median of the PMI scores. The median is chosen as a metric for the PMI scores to not have any outliers in the data bias the center point, as would happen with calculating the average. The values are taken from the sorted PMI scores following a formula for the total amount of associated descriptions n. The formula for $n = even$ is:

$$median = \frac{n+1}{2} \tag{4.1}$$

and the formula for $n = odd$ is:

$$median = \frac{\frac{n}{2} + (\frac{n}{2} + 1)}{2} \tag{4.2}$$

The most striking difference is seen in the difference in mean for $w = Nederlander$, where the median PMI score for the SRL pipeline is $1,865$ and for the syntax pipeline $-0,181$. This would mean that the descriptions extracted for SRL portraits generally have a positive association with the target word, whereas for the syntax pipeline this would be a neutral association, or no association in particular. This difference indicates that the syntax pipeline not only extracts more descriptions, but also more descriptions that are generally less associated. In other words, they co-occur less frequently than would be expected for them co-occur by pure chance. The juxtaposition of the two pipelines and their PMI scores would then indicate that the SRL pipeline extracts descriptions that would have higher associations with a query entity than the syntax pipeline would.

4.2.2 Qualitative Level

The qualitative level further investigates how the individual portraits differ from one another. It will discover how the descriptions differ qualitatively for each pipeline, and if and how the extracted information changes. To find any patterns that can provide an answer to the research question, this part will be guided with the following two underlying questions:

1. Are there any instances where the SRL predicted label is more accurate than the heuristics label?

2. Are there instances where the SRL predicted label provides more information of a higher granularity than the heuristics label?

The descriptions are compared against manually annotated descriptions. Further details about the annotation were discussed in section 3.2.2 and in Appendix 6. The article that has been manually annotated exists of letters sent by readers of the newspaper AD. The manual annotations focus only on the content of each letter, and omit the signed names and instructions to send letters. The manual annotation extracted 89 portraits, whereas both pipelines output 131 portraits for the same text. This is due to the manual annotation selectively annotating entities, and does not annotate last names, demonstrative pronouns and dates. Since the total amount would then only be 89 portraits, it would not yield generalizable results for calculating precision, recall and F1 scores.

Pipeline	Total portraits	Avg descriptions
Syntax	131	8.93
SRL	131	3.95

Table 4.9: Metadata manual annotations

Table 4.9 shows some metadata to support the earlier findings from the PMI scores. The total amount of portraits are 131 for both pipelines, yet the average amount of descriptions per portrait is 8.93 for the syntax pipeline compared to 3.95 for the SRL pipeline. This supports the observation from Section 4.2.1 that the SRL portraits contain significantly less descriptions than the syntax pipeline extracts. The rest of this section will investigate the implications of this difference. Example sentences from the manually annotated articles are presented together with the extracted portraits from both pipelines. Table 4.10 shows the first observations that further explain this.

"Betere zorg voor ouderen stemt mij optimistisch" *"Better care for the elderly makes me optimistic"*						
	SRL		**Syntax**		**Gold**	
Portrait ID	**Relation**	**Description**	**Relation**	**Description**	**Relation**	**Description**
ID1	label	zorg	label	zorg	label	zorg
ID1	property	goed	property	goed	property	goed
ID1	property	voor ouderen	property	voor ouderen	property	voor ouderen
ID1	A0	stemmen	property	optimistisch	A0	stemmen
ID2	label	mij	label	mij	label	mij
ID2	A1	stemmen	undergoer	stemmen	A1	stemmen
			undergoer	zorg	property	optimistisch
			undergoer	goed		
			undergoer	voor ouderen		
			undergoer	optimistisch		

Table 4.10: Comparison of extraction of microportraits

Table 4.10 shows the nanoportraits extracted from the following sentence:

(2) Betere zorg voor ouderen stemt mij optimistisch.
Better care for the.elderly makes me optimistic.
'Beter care for the elderly makes me optimistic'

The sentence is shown at the top of the table, providing the Dutch sentence followed by an English translation. The most left column shows the portrait ID, and then the columns for SRL, Syntax and Gold show the descriptions (labels, properties and/or roles) that each of them collected for the respective portrait.

The SRL pipeline and the gold annotations are quite similar, whereas the syntax pipeline differs the most. First, this can be seen for portrait ID1, where the gold annotation and the SRL pipeline both extracted *zorg* (care) as the phrase that has the A0 role for the verb *stemmen* (make). The syntax pipeline did not seem to pick up the role this phrase takes on for the verb *stemmen*. Both pipelines did successfully retrieve

the `undergoer` or A1 role for *mij* (me). This example then shows that the SRL pipeline is slightly more accurate in extracting roles and behaviors for phrases.

When it comes to extracting properties,the syntax pipeline includes *optimistisch* (optimistic) as a property for the label *zorg* (care) in ID1. However, *optimistisch* is a modifier of the label *mij* (me), which is represented in portrait ID2. Instead, the syntax pipeline represents *me* to have an `undergoer` relation with the property *optimistic*, just like every other word in the sentence. This is one reason why the average descriptions per portrait are much higher for the syntax pipeline than they are for the SRL pipeline. *Optimistic* is properly captured as property in ID2 for the gold annotations, however, the property is missing in the SRL output. It should be noted that the SRL system did predict the label AM-MNR for the word *optimistic*. Following the PropBank guidelines, manner modifiers specify how an action is performed and should be used when an adverb could be the answer to a question starting with *How?* (Bonial et al., 2012), which makes AM-MNR the right prediction here. Unfortunately, the current logic of the Microportrait Pipeline does not pick up these modifier arguments, unless the modifier is a label itself. This means that both pipelines do not score full points on this matter, as the SRL pipeline misses out on extracting argument modifiers that are informative for the microportrait and the syntax pipeline extends roles to each word in the sentence which can create noise in the resulting analysis of the microportraits.

This first example then illustrates a few differences already between the two pipelines compared to one another and the gold annotations:

1. The syntax pipeline omits some roles.
2. The syntax pipeline can connect properties to the wrong label.
3. The syntax pipeline extends semantic roles to all words in the sentence, instead of only the verb.
4. The SRL pipeline does not extract all properties in case they are an argument modifier.

These observations are further investigated in the following examples. First, the pattern of how roles and behaviors are extracted is explored, followed by how each pipeline handles copula constructions, then how each pipeline handles argument modifiers and properties, as well as some challenging verbs.

"... dat wereldwijd 50 miljoen moslims bereid zijn geweld te gebruiken"
"... that worldwide 50 million muslims are prepared to use violence"

	SRL		**Syntax**		**Gold**	
Portrait ID	**Relation**	**Description**	**Relation**	**Description**	**Relation**	**Description**
ID1	label	moslim	label	moslim	label	moslims
ID1	A1	zijn			property	50 miljoen
ID1	A0	gebruiken			A1	zijn
ID1					property	bereid
ID1					A0	gebruiken
ID2	label	geweld	label	geweld	label	geweld
ID2	A1	gebruiken	undergoer	gebruiken	A1	gebruiken

Table 4.11: Comparison of Roles and Behaviors between the Pipelines

Table 4.11 shows the portraits for the following subordinating sentence:

(3) dat wereldwijd 50 miljoen moslims bereid zijn geweld te gebruiken.
that worldwide 50 million muslims prepared are violence to use.
'that worldwide 50 million muslims are prepared to use violence.'

This table illustrates an issue that comes up when basing roles and behaviors on dependency information. How a certain phrase is parsed can lead to a certain way of extracting labels, properties and roles, which is bound by how its dependencies are parsed. Basing the extraction of descriptions on semantic representations can lead to different content of the microportraits. That is, by answering the question *who did what to whom?*, an SRL system can answer the *who* more accurately then a syntactic parse could. This is further explained for the example where the syntax-pipeline does not extract any roles for the label *Moslim* (Muslim) for ID1 in Table 4.11. In order to do this, we will inquire how the phrase *50 miljoen Moslims* (50 million Muslims) is parsed in the dependency layer of the NAF file. Figure 4.2 shows the relations between each token as parsed by the Alpino parser, in a simplified representation using the *head* → *dependency* representation following the Universal Dependency guidelines.

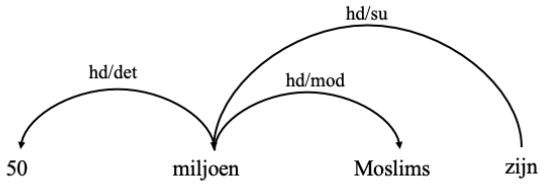


Figure 4.2: Dependency tree for the phrase "50 miljoen Moslims zijn"

The phrase *50 miljoen moslims* is parsed to have *miljoen* (million) as the head of the phrase, *50* as its determiner, and *Moslims* as its modifier, as illustrated in Figure 4.2. By having this syntactic representation, a dependency-based heuristic would not extract any modifiers or roles for the entity *Moslims*, as this token itself is a modifier of the head *miljoen* and as a result the 'lowest' node in the tree. Subsequently, the syntax pipeline does not see *Moslims* to be a participant for the verb *zijn* (to be), nor does it take on a role for the infinitive verb *gebruiken* (to use) at the end of the sentence. It only extracts *Moslims* as a stand-alone label. This results in the verb *gebruiken* only having an undergoer-participant, without anyone that *does* the violence. In other words, for the question *who did what to whom?*, the *who* is missing. In contrast, the SRL model labeled *Moslims* as A0 for the verb *gebruiken*. In this sentence *Moslims* is namely the semantic argument for the verb *gebruiken*, despite it not being its direct syntactic subject. This then does provide the agentive counterpart to the `undergoer` role for the verb 'to use'. This difference in depending on syntactic representations compared to semantic guidance results in a more accurate representation of roles and behaviors as extracted by the SRL pipeline.

Furthermore, Table 4.11 also shows how the pipelines differ in handling copula constructions. The gold annotation for ID1 also extracted *bereid* (prepared) as property for *Moslims* in the copula construction with *zijn* (to be). The syntax pipeline extracted *bereid* as a property in the portrait for *50 miljoen* which is not shown in the table, but

makes sense given its dependency structure assigning *miljoen* as the subject of the verb *zijn*. However, *bereid* is not a part of the SRL portrait, despite the SRL model predicting the A2-comment role for *bereid* , with *Moslims* being assigned the A1-topic role for the verb 'be' as a copula. *Bereid* would then only be part of a microportrait if it is extracted as a label itself. Unfortunately, since *bereid* has the part-of-speech tag *adjective* and has the dependency of a predicative complement, the SRL pipeline would not be able to extract this property as it would not be extracted as a label on its own. This pattern occurs more often within the SRL pipeline, when arguments are not a label themselves yet do provide meaningful information, which will be further discussed for Table 4.14.

"...dan leer je wel dat de multiculturele samenleving een immens drama is"
"...then you will learn what a great drama a multicultural society is"

	SRL		**Syntax**		**Gold**	
Portrait ID	**Relation**	**Description**	**Relation**	**Description**	**Relation**	**Description**
ID1	label	de	label	de	label	samenleving
ID1	label	samenleving	label	samenleving	property	multiculturele
ID1	property	multicultureel	property	multicultureel	property	immens drama
ID1	A1	zijn	property	drama	A1	leren
ID1			property	een		
ID1			property	immens		
ID2	label	een	label	een		
ID2	label	drama	label	drama		
ID2	property	immens	property	immens		
ID2	A1	leren				
ID2	A2	zijn				

Table 4.12: Comparison of how each pipeline handles copula constructions

Nonetheless, it does happen that both pipelines successfully retrieve all the information present in a copula construction, yet represent it in different ways. Table 4.12 shows the difference between the two pipelines in how they represent predicative complements. This table shows the portraits for the following sentence:

(4) dan leer je wel dat de multiculturele samenleving een immens drama
than learn you indeed that the multicultural society a great drama
is.
is.

'then you will learn what a great drama a multicultural society is.'

To assign semantic roles to this sentence, the SRL pipeline follows the PropBank guidelines for annotating *to be* as a copula, of which the frameset is shown in Table 4.13 below. The SRL pipeline then extracts the topic of the copula in ID1 and the comment of the copula in ID2, since that is the portrait that is created when the comment itself is extracted as a label.

be.01	
Role	**Description**
ARG1-PPT	Topic
ARG2-PRD	Comment

Example: The selection [ARG1] is better [ARG2].

Table 4.13: Frameset for `be.01`.

The syntax pipeline, on the other hand, has the heuristic that does not extract any agent or patient relations in case of a predicative complement, but rather extracts it as a property. That results in the label having no role in the copula 'event', but rather in having the predicative complement be included directly within its portrait. This can be seen in ID1 in Table 4.12, as *een immens drama* (a great drama) is part of the portrait as a property. The gold annotations do not seem to follow the PropBank guidelines in Table 4.12, which can also be seen in the gold annotations of Table 4.11

The gold annotations seem to favor the dependency based heuristic, by assigning *multicultureel* (multicultural) and *immens drama* directly as a property to the label *samenleving* (society). As a result, this phrase is then immediately assigned the A1 role for the verb *leren* (to learn), despite the whole clause strictly taking on the A1 role and not only the label *samenleving*. It appears that the SRL pipeline solved this by assigning *drama* the A1 role for *leren*. This then also immediately shows how the annotation task can be further improved for consistency in its guidelines, as discussed in Section 5.2.1.

As for interpreting the portraits after the extraction of copula constructions, it does not matter as much how they are represented in terms of accuracy and/or informativeness, as long as it is known what the structure is. Nonetheless, for such nanoportraits that stand on their own, the syntax pipeline is efficient to have all information about one entity in the same portrait. However, when a coreference system is able to combine these nanoportraits for each entity, the SRL pipeline would yield the same results, given that the coreference system takes into account that the A1 and A2 roles of the same copula belong to the same entity.

ID1: "Wij weten al jaren..."
ID1: "We've known for years..."
ID2: "de Islam steeds meer voet aan de grond krijgt, ook in Nederland."
ID2: "the Islam is increasingly gaining ground in the Netherlands"

	SRL		**Syntax**		**Gold**	
Portrait ID	**Relation**	**Description**	**Relation**	**Description**	**Relation**	**Description**
ID1	label	jaar	label	jaar		
ID1	property	al				
ID1	AM-TMP	weten				
ID2	label	Nederland	label	Nederland	label	Nederland
ID2	AM-LOC	krijgen	in-rol	grond	property	ook
ID2			in-rol	voet aan de grond krijgen	A2	voet aan de grond krijgen

Table 4.14: Comparison of how each pipeline handles modifying arguments

Table 4.14 shows the portraits for the following sentence, where the sentence represented in ID1 is glossed in (5) and the sentence which is represented in portrait ID2 is glossed in (6):

(5) Wij weten al jaren ...
We know for years ...
'We have known for years ...'

(6) ... de Islam steeds meer voet aan de grond krijgt, ook in Nederland
... the Islam always more foot on the ground gets, also in the.Netherlands
'the Islam is increasingly gaining ground in the Netherlands'

Next up, the pipelines extract argument modifiers differently, which was first discussed in Table 4.10. This is further illustrated in Table 4.14, which shows some cases where the argument modifiers (AM) have been extracted successfully as part of the portraits. These cases show that they become a portrait if the token that has the AM prediction is assigned its own portrait. For the first portrait, *jaren* (years) takes on the AM-TMP role for the verb *weten* (to know), which is then extracted as a portrait. Similarly, for the verb *krijgen* (to get), the location where something 'is gotten' is *Nederland* (the Netherlands). The syntax-equivalent of AM-LOC is the `in-rol`, where the event 'to get' happens *in* the Netherlands. Whereas the `in-rol` in combination with the label *Nederland* indicates that we are dealing with an adverbial of place, the `in-rol` can also be an adverbial of time or condition. The AM-LOC relation leaves no doubt what kind of adverbial we are dealing with. As such, the AM predictions of the SRL pipeline are then more informative, if it is connected during the coreference resolution to the right agent for the verb that is modified.

However, the difficult aspect of the second extracted portrait in Table 4.14 is that it uses figurative speech. The idiom *voet aan de grond krijgen*, which translates to 'gaining ground', does not have the same meaning as the verb *krijgen* (to get) on its own would have. A direct translation of *krijgen* would be 'to get', whereas the PropBank frameset

of 'gain' would be more appropriate for its non-compositional meaning. That would give *Nederland* the A2 role, which is the source something is taken from, rather than just the location. However, such idiomatic expressions are a known challenge within many fields of linguistics, and are difficult to train a machine learning model for if not for sufficient data.

"want ook Nederland Islamiseert." *"since also the Netherlands becomes more Islamic."*						
	SRL		**Syntax**		**Gold**	
Portrait ID	**Relation**	**Description**	**Relation**	**Description**	**Relation**	**Description**
ID1	label	Nederland	label	Nederland	label	Nederland
ID1	property	ook	property	ook	property	ook
ID1	A0	islamiseer	agent	islamiseer	A1	islamiseren

Table 4.15: Comparison of an Error and Flaw

Table 4.15 shows the difficulty that both the SRL model and the dependency-based heuristics seem to struggle with. This is shown for the following sentence:

(7) want ook Nederland Islamiseert.
since also the.Netherlands Islamifies
'since also the Netherlands becomes more Islamic'

The verb *islamiseren* (Islamify) is a process of change that one undergoes, which makes the subject of the verb a patient rather than an agent. This can then be seen as an error of the SRL model, and not of the pipeline itself. For the portrait of the syntax pipeline in Table 4.15, this is a flaw within the dependency heuristics. The mistake that was made by the SRL pipeline could then be resolved by sampling specific data to further finetune the model on, whereas the flaw within the dependency heuristics cannot be as easily resolved.

4.2.3 Conclusion

In conclusion, the evaluation of the Microportrait Pipeline on a quantitative level is further investigated on a qualitative level. The associations that came out of the PMI scores are further explained by the qualitative evaluation. That is, the qualitative analysis explained how the syntax pipeline extracts many more descriptions, as well as that it extracts more syntactically related terms. Furthermore, the combination of both evaluations explained the differences between the two pipelines, as well as how the results change now that SRL has been implemented. The two guiding questions posed at the start of this chapter then helped in formulating how the pipelines differ from one another:

1. Are there any instances where the SRL predicted label is more accurate than the heuristics label?

2. Are there instances where the SRL predicted label provides more information of a higher granularity than the heuristics label?

What the results have shown is that the syntax pipeline extracts more descriptions for each portrait, for example because the syntax portraits can extend a semantic role such as `undergoer` to each word within the phrase. In contrast, the SRL pipeline only connects the semantic role to the verb it is related to and in turn creates better generalizations when assigning roles. This higher amount of generalization in the output of the SRL pipeline creates less noise when interpreting the portraits, and avoids the inclusion of descriptions within a portrait that do not belong, as seen in Table 4.10. However, because of this generalization it can also lose information, as extensively discussed for Table 4.7. In contrast, the syntax pipeline does include this information at the expense of being noisily represented with wrong relations. Nonetheless, it can be questioned which is more desirable.

It also appears that roles and behaviors in the SRL pipeline are more accurately extracted than the syntax portraits, as some of the semantic roles are not represented in the syntax portraits. Lastly, argument modifiers can provide more precise information within a portrait then the prepositional roles, yet is not always extracted. Overall, the results section has shown indications that the pipeline supplemented with SRL information can provide more accuracy and informativeness, although it is not flawless yet.

Chapter 5

Discussion

This thesis updated the Microportrait Pipeline to be able to extract SRL information for its portraits, rather than basing this on syntactic heuristics. This change was evaluated on a quantitative level through exploring differing patterns in highly associated terms through PMI scores, as well as on a qualitative level by exploring differences between how individual nanoportraits are made up.

Section 5.1 will answer the research question by discussing the main findings and their implications. The thesis will be further scrutinized by discussing the choices that were made and what limitations they come with in Section 5.2.1, and lastly, an outlook on what is still there on the horizon is presented in Section 5.2.2.

5.1 Research Question and Answers

The ultimate goal of updating the Microportraits is to improve the information that the pipeline presents. This then begs the question what we want to know when applying the technology of Microportaits. Broadly speaking, the pipeline based on syntax gives a representation of terms that are syntactically related to an entity, whereas the iteration of the pipeline that includes SRL changes this more towards a semantic representation of what a given text conveys. In other words, it is more fit to answer the question of *who did what to whom?* Now we can move to answer the research question of this thesis, which was stated as:

> *To what extent can the inclusion of semantic role labeling in the microportrait extraction pipeline increase the granularity and improve the accuracy of microportrait extraction in Dutch News?*

It is now possible to answer this question, given the findings presented in Chapter 4. The main findings are presented here and related to the research question. Starting with the first and foremost finding:

1. The syntax pipeline extracts significantly more descriptions, meaning that the SRL pipeline misses certain descriptions.

The syntax pipeline creates portraits that are on average double the size than the portraits of the SRL pipeline. This indicates that the SRL pipeline creates more selective semantic portraits that focus more on finding an answer to the question: *who did what to whom?* The portraits produced by the syntax pipeline then often include

information that is related through some syntactic construction, as illustrated with the PMI scores. This means that the SRL pipeline is better in making generalizations yet at the cost of not including some informative terms. However, as already discussed in section 4.2.1, the syntax-pipeline also does not represent these terms accurately, assigning the label of a portrait a semantic role for a noun that occurs in an adjunct. Between the current two pipelines, this means that a trade-off exists, between wanting as many descriptions as possible with the result of them containing quite some noise and wanting more selective semantic representations at the cost of missing out on some descriptions that can provide more information.

The next finding also relates to the question *who did what to whom?*, and why the SRL portraits are better fit to answer this question.

2. Roles and behaviors in the SRL portraits are more accurately extracted than the syntax portraits.

This relates to two findings, namely first that the SRL portrait finds the *who* regardless of the dependency structures in a sentence or phrase, and second that the SRL portraits more often capture roles within a portrait that are overlooked in the syntax portraits. This result should be countered with the fact that the SRL model also predicts false positives and as such does not assign the right label in every scenario. Nonetheless, for many verbs that are overlooked in the syntax pipeline, such as end-of-sentence infinitives, the SRL pipeline outperforms the syntax pipeline. Both findings 1. and 2. point towards an improvement of the pipeline with SRL. Especially as the goal of the pipeline is to create microportraits about entities that can bring patterns to light about biased choices in language, which would benefit from an accurate semantic representation that answers the *who did what to whom* question.

3. Argument modifiers extracted by the SRL pipeline are more informative, if extracted.

The modifying arguments that are part of the SRL portraits provide more information on what the function of that argument is. These arguments are extracted by the syntax pipeline based on the preposition that precedes the phrase, providing `in-roles`, `with-roles` and `to-roles`. However, these prepositional phrases can provide information regarding time, location, condition and whatnot. The AM-LOC or AM-TMP predictions in the SRL portraits immediately provide this information. However, these modifying arguments are not always included in the SRL portraits, only if that word is extracted as a label itself. The fact that many modifying arguments are not extracted is because the pipeline focuses on entities rather than events, whereas SRL is based on events and does not base its predictions on the entities. This would then be something to take into account when combining the nanoportraits through coreference resolution. Namely that some portraits are connected through an event rather than an entity. Despite this being an improvement, it is still subject to further development.

5.2 Reflections and Looking Forward

5.2.1 Reflections

As is inevitable when creating and executing a research design choices have to be made. Whether those choices are the most accurate for the research, is something that should

be reflected on. Namely, why has a certain choice been made, what are its implications and does it come with any limitations to the research outcome? As such, this section will reflect on some of those choices and how it could have possibly influenced the research outcome. The core of this thesis is the implementation of an SRL system, and several choices have been made throughout. One of them is to use PropBank as the semantic database to train the system on rather than FrameNet. Second, the system is trained on only verbal predicates, and misses out on non-verbal predicates, and the system is trained on Dutch and English UP data, which differs from the data that it is applied on. Also, the choice was made to use a computationally complex SRL system, whereas a traditional model could have been trained.

PropBank versus FrameNet

This SRL system has been trained on PropBank annotated data, which has been chosen because of 1) its easy availability and it contains large datasets in multiple languages, 2) its consistency in labels that makes the training process and predictions more efficient and 3) given its basis in Proto-Agent and Proto-Patient roles that avoids the fuzziness of many different semantic roles. However, it is also a possibility to use FrameNet for identifying roles in microportraits, and there are several reasons that make FrameNet a good fit.

The first reason is based in the underlying theory of Frame Semantics. Frame Semantics is developed by Fillmore (1976) to rely more on contexted experiences to define the meaning of words. This takes cognitive processes into account as the interpretation of language depends on our perception of the context as well as how we recall any earlier experiences with a certain utterance (Fillmore, 1976). This relates to the process of categorization that also lies at the basis of assigning characteristics to a stereotype, as discussed in Section 2.1. By using a theory of semantics that aligns the cognitive processes that are at play when using language as well as when assigning stereotypes, the subsequent reasoning about the microportraits would be more in line with how language is produced and interpreted. That is, frame semantics is based on frames, which can be seen as structured background knowledge that language-users use to interpret linguistic expressions (Fillmore, 1985). Each word, mostly predicates, evoke these frames that relate to a certain scene that organises the roles, relations and expectations we have (Fillmore, 2006). As such, put simply, when we hear the verb *to kill* in a sentence, we do not expect an ARG0 - ARG1, agent - patient relation, but rather we expect a killer and a victim. These same roles of killer and victim exist in the framesets for the verbs *assassinate, murder* and *suffocate*. These roles then provide more detailed information on which roles and behaviors entities take on in a sentence, which can improve the interpretation of microportraits compared to agent-patient, Proto-Agent - Proto-Patient, ARG0-ARG1 roles.

That leads to the second reason why FrameNet would be a good fit, as it combines verbs that yield the same roles, relations and expectations under the same frames. FrameNet organizes verbs that take on the same arguments in a sentence and can be grouped within the same rolesets, whereas PropBank defines these rolesets for each predicate separately. That is, each roleset in PropBank further specifies the A0 and A1 roles to be a *giver* and *thing given*. These argument specifications exist for each verb, creating rolesets for every defined predicate. Such a verb-specific approach from PropBank comes with its limitations (Palmer et al., 2010). First, since each role label is verb-specific, it is not possible to make generalizations across verb classes. Each role

label is further specified for each verb to have its own frameset, which does not make it possible to determine when two verbs' arguments have the same role (Palmer et al., 2010). Second, if the goal is to train a machine learning model on each specific frameset, this would become a training task with huge amounts of data. That is why this verb-specific approach that labels each predicate's arguments with its further defined roles is ignored in most SRL models, similarly to the one trained in this thesis, where A0 and A1 are related to Proto-Agent and Proto-Patient roles. Given that these two labels take up the vast majority of the arguments, it is an effective approach to ignore the detailed rolesets and only perform the task with the general A0, A1, etc. roles. (Palmer et al., 2010). In contrast, a frame-specific approach as in FrameNet would allow a system to generalize over verbs that belong to the same frame, as the verb's arguments then take on the same roles.

As such, FrameNet is able to provide additional semantic information that can help in the creation of microportraits. Unfortunately, creating datasets to train, develop and test frame semantic parsers is a difficult task. One system that does this is the LOME parser (Xia et al., 2021), that also exploits cross-linguistic patterns. However, besides the fact that LOME does not function for Dutch, the implementation of the LOME parser is not as straightforward as training an SRL system on PropBank. For example, the LOME parser is not readily available on Huggingface as different BERT models are.

Predicate Identification

The task of an SRL system that identifies and classifies semantic arguments given a predicate obviously relies on which predicate is identified. The current system is not first trained on predicate identification and classification, nor has a different system aimed at this specific task been developed. Rather, predicates have been identified based on their part-of-speech tag in their NAF files, where only verbs are extracted. This comes with two implications.

First, this means that only verbal predicates are included in the system to predict its semantic arguments for. However, adjectival predicates also occur quite often in Dutch (and other languages) in copula constructions. As already mentioned in Section 3.1.2, copula constructions are extracted as properties connecting to the subject of that sentence. That is because a copula construction does not strictly require an agentive relation. As for the SRL model, this is accounted for that the *to be* verb in copula constructions is annotated as such. Nonetheless, it does run the risk on missing out on certain non-verbal predicates and as such omitting possibly useful information for the microportraits.

Furthermore, by not using a system that identifies and classifies predicates, the system has no information on predicate-sense disambiguation. For the system itself, this can result in a difficulty to disambiguate polysemous verbs. However, the proto-agent and proto-patient roles often generalize over polysemous verbs. For example, in Dutch *houden* can be used in the sense of *holding a presentation* and *loving someone*, yet the A0 and A1 roles that are assigned to both instances would be the same participants in the sentences:

Ik[A0] hou van jou[A1]

Ik[A0] houd een presentatie[A1]

SRL Model Architecture

The model architecture is based on several decisions. By comparing three different finetuned models, it was possible to decisively choose the XLM-RoBERTa model that has been finetuned on both English and Dutch data, given that it slightly outperforms the other two models. However, a choice does not have to be based only on the model's performance, given that the XLM-RoBERTa model is more complex in its architecture and takes up more computing power to finetune and use than for example BERTje. So the question then is whether this increased computing power is worth it given the slightly better performance. There are two reasons for this, as 1) for a delicate task such as microportrait extraction of which the output is used to reason over microportraits, it is desirable to have the most accurate predictions to reduce the chance of wrongly reasoning over the patterns that occur, and 2) given that they are both transformer-based models, the relative difference in computing power is not much, given the amount of time it took to train the models was very similar. If it were the case that a traditional machine learning model based on feature engineering yielded the same results as the BERTje model, a case could have been made to use that model as the default. Nonetheless, given the complexity of the SRL task, the time it takes to feature engineer a model and the limited time to execute this study, a traditional ML model has not been created for this thesis.

Head word labeling

The SRL system is trained for identifying and classifying the head word of an argument, and so is the pipeline with SRL built to extract only the heads of a semantic argument and not the entire phrase. Even though it has been argued in Section 3.1.2 why this is the case based on XLM-R's and the pipeline's architecture, it also comes with its limitations. When only labeling the head of a phrase, all modifiers within the phrase are lost. Adjectives that describe the head word are not included in the labeling of the semantic role, neither are appositions or modifying clauses. By making this choice, the pipeline itself should extract all these extra descriptions, yet unfortunately does not always succeed in doing this or uses shortcuts that results in noise. It can then be considered to not only label the head word, but the whole span in order to improve the extraction.

Furthermore, the PMI scores showed how different labels can still one of the most associated terms for a label about which a portrait is made. This extraction is based on the heuristic that assigns appositions, multi-word phrases and determiners labels within a portrait. It can be the case that these parts of speech indeed provide meaningful information, yet they can also provide unnecessary information, such as the time indication as shown in Table 4.5. This could potentially be resolved by extending the SRL task to be span based, which would then include appositions and longer phrases as one semantic role. The SRL system would ideally be able to filter for relevant parts of a phrase to include as part of the semantic role.

Evaluation

The PMI scores that have been calculated also include negative scores, which are unobserved co-occurrences that can lead to PMI-scores that continue to negative infinity (Salle and Villavicencio, 2023). These negative values tend to be unreliable, unless we would have an enormous, say infinite corpus. Finite corpora yield unreliable statistics

(Salle and Villavicencio, 2019), and evaluating such infinite negative scores of unrelatedness with human judgment would be a near impossible task (Jurafsky and Martin, 2026). Nonetheless, this issue is most often cited in the literature of distributional semantics for creating reliable static word embeddings. For the purpose of this thesis it was to find the most associated terms with a keyword, and we are not further interested in establishing relatedness between the words themselves. As a result, changing PMI to PPMI by clipping each negative value at 0 would not change the results. It would only result in having a 0 in Table 4.8 instead of the only reported negative score of $-0,181$.

As for the manual annotation, clear guidelines should be set if the goal is to calculate standard evaluation metrics. First, it should be established which labels are to be retrieved by the mannual annotation. If precision and recall are to be calculated, the manual annotation should also retrieve all labels to create a relevant gold dataset. However, as already established in Fokkens et al. (2018), this makes the annotation task more difficult than by only annotating entities of interest. Furthermore, the two pipelines now handle copula constructions differently. The syntax pipeline assigns the predicative complement in a copula construction as a property of the label, whereas the SRL pipeline creates a new portrait for the predicative complement and assigns it the A2 role as set out by the PropBank annotation guidelines. As discussed in section 4.2.2, a coreference system can combine the separate portraits of the SRL pipeline into one, which would make the PropBank annotation guideline a more straightforward annotation rule. However, the rule to assign the predicative complement as a property may be more intuitive for annotators.

5.2.2 Looking Forward

In addition to reflecting on what has been done, it is also possible to look forward. As currently the pipeline has no working coreference system, is not robust for multilingual applications and could be able to include more linguistic information. This section will further explore what possibilities are still at the horizon for the Microportrait Pipeline.

First of all, as the current pipeline only outputs nanoportraits on a sentence level, a state-of-the-art coreference system would greatly improve the functioning of the pipeline. This has already been indicated in Chapter 4, where both the A1 and A2 roles that both belong to a copula as well as all argument modifiers predicted by the SRL pipeline can then be connected to its respective entity. A coreference system would then not only create microportraits for each entity in a text, but also immediately improve some limitations that are identified for the SRL system.

Second, there are more linguistic structures that can be extracted from text to capture more implicit stereotypes on top of the more 'overt' language use that is extracted in labels, properties and roles. As such, drawing from Schmeisser-Nieto et al. (2022), different linguistic structures can be included to account for more implicit ways of communicating stereotypical behavior. One of them can be found in habitual aspect, where this aspect property characterizes a specific event or action as the default behavior of the target group (Schmeisser-Nieto et al., 2022).

Furthermore, ideally the pipeline would not only function for Dutch, but would be applicable to a plethora of languages. Perhaps not all 7.170 languages,[1] yet an effort can be made to include more than one. One effort has already been made to update

[1] https://www.ethnologue.com/insights/how-many-languages/

the pipeline to Polish, which resulted in a new pipeline having to be built from scratch due to the differences in Polish and Dutch NLP tools (Kluska, 2023). Another effort that can be made is by updating the pipeline to be more robust for current parsers. As the pipeline bases its heuristics for sentence analysis on the annotation scheme in the Alpino Treebank, which is based on Head-Driven Phrase Structure Grammar (HPSG) (Kakkonen, 2006), the pipeline is not adapted for current parsers that are being developed that base their dependency relations on Universal Dependency (UD) relations (De Marneffe et al., 2021). As such, the UD equivalents of the HPSG dependency labels can be added to the heuristics to make the pipeline more robust for different parser information. The equivalence of these labels can be derived from the Lassy syntactic annotation guidelines (van Noord et al., 2010), as well as the UD syntactic annotation guidelines. [2] Furthermore, their equivalence should be based on the information that each label provides. For example, in copula constructions Lassy annotates the verb as the root and the complement as the predicate `predc`, whereas UD annotates the non-verbal predicate as the root and the verb as a copula. This requires careful consideration of which syntactic information is stored under which labels.

As a last point, the different applications of the pipeline can also be further extended than only extracting microportraits from articles. Eventually, when the pipeline is void of its current flaws, its application can reach further than only investigating representations in the media. Since it is a system that extracts information directly from natural language, this can also be applied to social media, law documents, police records and other genres. Since this method is stereotype- and genre-agnostic, the application of the pipeline should be manifold.

[2] https://universaldependencies.org/guidelines.html

Chapter 6

Conclusion

This thesis investigated to what extent the inclusion of semantic role labeling can increase the granularity and improve the accuracy of microportrait extraction in the use case of Dutch news. As such, a BERTje model and two XLM-RoBERTa models have been finetuned which resulted in one best performing model to be used for this thesis. The XLM-RoBERTa model that was finetuned on both Dutch and English PropBank data was implemented into the Microportrait Pipeline. This replaced the dependency-based heuristics for extracting roles and behaviors with semantic role predictions.

The results were able to provide useful insights to form an answer to the research question:

> *To what extent can the inclusion of semantic role labeling in the microportrait extraction pipeline increase the granularity and improve the accuracy of microportrait extraction in Dutch News?*

The findings show that the SRL model achieved strong performance on the most frequent argument roles and that the resulting SRL-based microportraits differ from the syntax-based portraits on both a quantitative and qualitative level. The PMI analysis showed changes in the descriptions that are most strongly associated with target groups, while the manual analysis indicated that the semantic arguments generally provide more informative and semantically meaningful representations. Compared to the syntax-based heuristics, the SRL-based pipeline produced less noise and was better able to capture the roles and behaviors assigned to entities.

At the same time, several limitations remain. The current implementation still operates on sentence-level nanoportraits as it does not have a working coreference-resolution system implemented. Moreover, some SRL predictions such as argument modifiers are not yet consistently extracted and non-verbal predicates are not fully accounted for. Nonetheless, the results indicate that replacing syntactic heuristics with semantic role labeling constitutes a promising improvement of the Microportrait Pipeline.

By introducing more semantically grounded representations of roles and behaviors, this thesis contributes to the development of a transparent and stereotype-agnostic approach to studying stereotypes in natural language. Future work can further expand the pipeline with coreference resolution, extraction heuristics for labels and properties based on Universal Dependencies instead of Alpino labels and perhaps include more linguistic structures that can contain biases. By further developing the Microportrait

Pipeline, its application then does not have to be limited to news articles, but can be expanded to include any genre of interest.

Appendix

Appendix A

	Precision	Recall	F1	Support
A0	0.862	0.867	0.865	688
A1	0.804	0.864	0.834	1123
A2	0.704	0.770	0.735	370
A3	1.0	0.385	0.555	13
A4	0.666	0.5	0.571	8
AM-ADJ	0.0	0.0	0.0	70
AM-ADV	0.6	0.685	0.64	70
AM-CAU	0.466	0.636	0.538	11
AM-COM	0.0	0.0	0.0	2
AM-DIR	0.75	0.75	0.75	4
AM-DIS	0.757	0.858	0.804	113
AM-EXT	0.75	0.545	0.631	11
AM-LOC	0.525	0.673	0.590	92
AM-MNR	0.683	0.848	0.757	79
AM-MOD	0.928	0.966	0.947	148
AM-NEG	0.9	0.978	0.937	46
AM-PRD	0.5	0.6	0.545	10
AM-PRP	0.5	0.846	0.628	13
AM-REC	0.0	0.0	0.0	1
AM-TMP	0.822	0.875	0.847	344
C-A1	0.666	0.4	0.5	5.0
R-A0	0.833	0.952	0.888	42
R-A1	0.740	0.8	0.769	25
R-AM-LOC	0.833	1.0	0.909	5
R-AM-MNR	0	0	0	1
R-AM-TMP	1.0	0.857	0.923	7
_	0.988	0.980	0.984	28.348
Macro Average	0.632	0.650	0.631	31.579
Weighted Average	0.968	0.966	0.967	31.579

Table 1: Performance of XLM-RoBERTa finetuned on English and Dutch

Appendix B

177015365	177030227	177045525
177019423	177030255	177045892
177019724	177030291	177046086
177019789	177030346	177046150
177021542	177030537	177046171
177021712	177030710	177046344
177021902	177034525	177046441
177022632	177035965	177046484
177023249	177037637	177046549
177023973	177039152	177046819
177025236	177039864	177046887
177025284	177041005	177046915
177026158	177042741	177046940
177026701	177043450	177047030
177026861	177043731	177047053
177028193	177043822	177047121
177028556	177043902	177047436
177028986	177044193	177047981
177029005	177044266	177049186
177029733	177044722	177049382
177030105	177044817	177051005
177030157	177045228	177051208

Table 2: File ids with no SRL predictions

Appendix C - Data Statement

Following the proposal of Bender and Friedman (2018), this section will explicate the data that is being used in this thesis. This provides context to understand to what extent the results of the study can be generalized, how the technology can be appropriately implemented and what biases can sneak into the software at what moment (Bender and Friedman, 2018). This data statement should help mitigate any harmful social impact, such as exclusion, overgeneralization, and underexposure (Hovy and Spruit, 2016). Through highlighting the characteristics of the dataset, it will also allow reasoning about any possible effects of the chosen data, as well as further aid in generalizability and reproducibility.

Curation Rationale *Which texts were included and what were the goals in selecting texts, both in the original collection and in any further sub-selection?* The data is selected to create an overview of political news articles in the Netherlands. A distinction is made between political and non-political news, where news is political when one of the political parties, party leaders, parliament members, or one of the ministers is mentioned in an article. The articles were selected between the time period of September 5 2016 till March 15 2017. The different media outlets of which the data was selected can be found in 3 below.

The Telegraaf is often considered to be a more right-wing newspaper, where the *NRC* is considered to be liberal and in the center of the political spectrum. *Trouw* and

National Newspapers	Online Newssites	Blogs
Algemeen Dagblad	AD.nl	Joop.nl
De Telegraaf	Telegraaf.nl	Geenstijl.nl
NRC Handelsblad	NRC.nl	
Trouw	Trouw.nl	
De Volkskrant	Volkskrant.nl	

Table 3: Overview of Media Outlets

De Volkskrant are more left-wing newspapers. The *AD* profiles itself as a neutral news outlet. It should be noted that the data used for this research is relatively outdated, as it was collected from a time period almost ten years ago. The reason for this is to be able to compare the output of the syntax-based microportraits pipeline to the SRL-based pipeline. This thesis therefore does not claim to provide a reflection of the current status of stereotypes in Dutch news media.

Any subsets that are used throughout this thesis are created at random, meaning that the origin of the data was not a factor in the deliberation on how a subset should be created. As a result, it is unknown what the distribution of data outlets is in any subsets. However, since this thesis does not compare media outlets nor does it take their position within the political spectrum into account, this should have no effect on answering the research question.

Language Variety *Languages differ from each other in structural ways that can interact with NLP algorithms.* The pipeline and SRL model are both engineered for the Dutch language with BCP-47 nl-NL (Phillips and Davis, 2009). That means that the language up for research is the standard variation spoken in the Netherlands. Since this variation of Dutch that is spoken in the Netherlands is not a lesser-known, endangered language, nor a regional variety or a historical stage of the language, it is not necessary to further disambiguate the language variety (Gillis-Webber and Tittel, 2020). Nonetheless, it is recognized that this research is geared towards the standard variety of Dutch, and does not take any dialectal variations within the Netherlands and in Flanders into account.

Annotator Demographic *What are the demographic characteristics of the annotators and annotation guideline developers?* The data and the pipeline of this thesis has two stages of annotations. The first annotation stage is that of the SRL model. This uses SRL annotations that are converted from English to Dutch, of which the annotations are done using English SRL models that result in higher quality as measured on 3 TL language with gold SRL labels (Jindal et al., 2022). The automatic conversion to Dutch is then based on these English annotations. As far as I am aware, no gold annotations exist yet for the Dutch UP dataset. The second stage of annotations is that of the evaluation set of the microportraits pipeline. The subset that is used for evaluation has been annotated by me, a native Dutch speaker, with two Dutch speaking parents that speaks English fluently as a second language. Moreover, I am a student of computational linguistics, and know how linguistic structures work. No other annotators have engaged in this task of annotating, meaning there is no inter-annotator agreement that can verify the quality of the annotations or that of the guidelines. This is further discussed in section 5.2.1.

Text Characteristics *Both genre and topic influence the vocabulary and structural characteristics of texts, and should be specified.* Depending on a number of non-

linguistic factors, the choices of a language-user can change in morphology, word-choice and grammar (Biber, 1995). The register of the data is journalistic news articles, where register here means to refer to news articles as a variety that is associated with a particular situational context and purpose (Biber, 1995). The articles are selected on whether they are political in nature, which influences the vocabulary and how the texts are structured. Moreover, both articles from the paper as well as from their respective websites are retrieved, which can mean that the platform influences the language that is used within an article.

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